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# Determinants of Drug Prices: An International Benchmark Systematic Review

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# Determinants of Drug Prices: An International Benchmark Systematic Review

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# Abstract

**Objective**: In this international benchmark systematic review, we report on recent determinants of net drug prices in a national and international context to facilitate the regulation of net drug prices by policy advisors worldwide. Determinants of drug prices were divided into non-modifiable and modifiable and categorized as pertaining to a country's income level, pharmaceutical market system and its policies and government.

**Primary outcome**: net drug prices or drug price variance.

**Design**: We systematically searched PubMed, EMBASE, Web of Science and Cochrane Library for peer-reviewed articles published after 2004 that reported an association of the primary outcome with one or more determinants. We performed a best-evidence synthesis of these associations for determinants that were covered in at least three studies.

**Results**: 31 publications were included, of which only one publication described net drug prices and 30 described retail drug prices. Five modifiable determinants were associated with lower retail prices: generic market portion, discounts, tendering policies, central (governmental) purchasing and pricing regulation schemes. The originators market portion and a system in which mark-ups are common were associated with higher retail prices. Retail prices were highest in the United States, even compared to other high-income countries. A positive association between national income level and drug retail prices could not be established among middle- and high-income countries. Retail prices were highest in low-income countries when adjusted for purchasing power parity.

**Conclusions**: Literature on determinants of net drug prices is extremely sparse. Various healthcare system interventions, market-specific and governmental regulations are consistently associated with lower retail prices. Some of these interventions are easily implementable in developing or middle-income countries, such as tendering, central purchasing and fixed pricing regulation schemes. Additional net price benchmark studies are needed to overcome the lack of price transparency and to quantify the effectiveness of policy measures on net drug prices.

**300 words**


## Strengths and limitations of this study

- An international benchmark systematic literature review was conducted to assess all recently identified modifiable and non-modifiable determinants of net drug prices.
- Only one published study conducted determinant analyses with net drug prices, while the rest included retail drug prices.
- The structured best-evidence synthesis is able to quantify heterogeneous evidence and summarizes the association between drug price (variance) and the reported determinants.
- This study design only allows for correlative analysis and cannot establish direct causal associations between determinants and price (variance).

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# Introduction

The global average share of healthcare costs in gross domestic product (GDP) increased from 4.6% of GDP in 1970 to 9.0% of GDP in 2016. [1] Several factors contributed to this rise, such as ageing populations, technological advances, the development of (expensive) new drugs and increased insurance coverage. Consequently, countries worldwide face the challenge of decreasing this healthcare cost burden. A significant driver of rising health care costs is pharmaceutical expenditure, so interventions to decrease drug costs through lower drug prices are warranted. Pharmaceutical expenditure is expected to reach 1.5 trillion dollars in 2023. [2, 3] Currently, a lack of transparency due to contract and pricing agreements between purchasers and manufacturers of drugs limits insight into net drug prices, because price comparisons are based on retail prices. Net drug prices are the prices actually paid by the purchasing healthcare institute, while retail prices exclude various individual pricing agreements such as rebates. [4] The effects of various policy measures are, therefore, unavailable to most purchasers and policymakers, which makes the drug price market opaque. [5-8]

By benchmarking drug prices in different countries, determinants of drug prices can be identified. We make a distinction between determinants that are modifiable by policymakers and determinants that are not. Modifiable determinants could act as a target for potential interventions to decrease drug costs. Present drug policies are highly heterogeneous and have varying effects on drug prices. Therefore, identifying and harmonizing effective price management strategies may aid in accomplishing lower prices. However, international and national price comparisons of net drug prices (prices actually paid by the purchaser) are scarce in the literature. To our knowledge, the first publication that quantified drug price differences was performed in 1971 in the United States. [9] In this study, the net price paid by the US government was compared to the purchasing price of community pharmacies for the 50 most prescribed medications. Central governmental purchasing by the US resulted in a relative discount of nearly 50% over local pharmacies. Since then, research on drug prices (excluding price ratios) and, more specifically, net drug prices has been very limited, but in recent years some determinants of retail drug prices were identified. [5, 10-13] Most reported determinants are non-modifiable (e.g., geographical factors, GDP and product age). However, some are modifiable. In individual studies, systems allowing for mark-ups (e.g. processing costs or profit passed on to purchaser), degree of market competitiveness, the generics market proportion and degree of governmental intervention (reference pricing and education about efficacy of generics) have been found to influence drug prices in some countries. [14-19] A systematic analysis of drug price determinants that are published in benchmark studies is, however, missing.

Therefore, in this international benchmark systematic review we aim to identify determinants of net drug prices and more specifically modifiable determinants that may serve as potential targets for drug price optimization. In addition, we give an overview of the literature in drug price benchmarking and execute a best-evidence synthesis of all reported determinants of net drug prices.

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# Methods

This systematic review was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. [20]

## Patient and public involvement

There were no patients involved in this study.

## Search strategy and information sources

The databases of PubMed, EMBASE, Web of Science and the Cochrane Library were searched until July 2020. We aimed to investigate which determinants influence drug prices by using benchmark or comparison studies that measure at least one such determinant. All synonyms of the three topics of this study, *prices*, *drugs/medicines* and *benchmark*/*comparison*, were included in the search strategy - with the exclusion of *cost-effectiveness* - acknowledging the respective database formats. All peer-reviewed articles that reported at least net prices (excluding price ratios) in a benchmark or comparative study were included. Both national and international studies were included, as well as pharmacy or hospital comparisons, prospective, transverse and retrospective studies. Only articles published between 2004 until and including 2020 were included for two reasons. First, our aim is to investigate current determinants of drug prices. Second, articles on this topic published before 2004 were scarce and likely less relevant, as the pharmaceutical market underwent significant changes in the last fifteen years. [21] Articles were excluded if they only reported (a combination of) policies, efficacy, cost-effectiveness studies or studies examining general treatment costs (such as *diagnosis treatment combinations* [DBCs] in the Netherlands). We executed a two-way process of reference and citation checking of the included articles.

## Study eligibility and outcomes

The PICO (research format of population, intervention, control, outcome) consisted of drug or drug group (P), present determinant (I), non-present determinant (C), drug price or price variance (O). Therefore, all peer-reviewed studies that reported a measure of association between drug prices (meaning: not relative prices or price indices) or price variance (dependent variable) and one or more determinants (independent variable(s)) were included.

## Selection process

First, titles and abstracts (or if unavailable, full text) were independently screened by two researchers (JMJD and AdA). The full texts of potentially relevant articles were independently assessed for inclusion by two researchers (JMJD and AdA). Discrepancies between the researchers were resolved through discussion and consensus. If no consensus was reached, a third researcher (BvdB) was consulted.

## Data collection and extraction

A standardized extraction worksheet was used to collect study characteristics, methods and outcome information from all included articles by both AdA and JMJD. The extraction worksheet was ultimately cross-checked. The following items were collected: first author name, year of publication, manuscript title, country of study, study design, selection criteria (regarding included medications, institutes), (type of) price comparison/benchmark, studied determinants of drug prices and association between determinant and drug price.

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### Risk of bias in individual studies and across studies

Quality assessment of the included studies was done with the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies of the National Institute of Health (NIH). [22] As the main purpose of this study is to give an overview of the field and to conduct a best-evidence synthesis, this instrument was only used to assess the quality and characteristics of the included studies and not to exclude additional studies. This specific assessment tool was used, because studies were mostly cross-sectional, sometimes observational and generally heterogeneous. As this tool is only meant to guide researchers in assessing such article's strengths and weaknesses, potential consequences of the identified risk of bias were interpreted by three authors (JMJD, AdA, BvdB) and consensus was reached based mainly on potential consequences on research results, as is also recommended by the NIH. In compliance with the assessment tool methods, studies were categorized in three quality levels: Good (all relevant qualifications met), Fair (1 or 2 not met) or Poor (3 or more not met).

Within the studies, selective reporting was assessed by carefully exploring the entire results section and supplementary material for non-reported associations.

### Summary measures and evidence synthesis

A qualitative best-evidence synthesis was performed to identify determinants of drug prices and variance in drug prices. Drug price variance reflects the potentiality of price differences of the same drug arising with comparable exposure to the analyzed determinant. Unfortunately, the heterogeneity between studies meant a quantitative analysis was not feasible. Determinants measured in every study were counted. If a determinant was studied three times or more, then it was analyzed and a best-evidence synthesis was conducted. If the determinant was studied once or twice, it was only shortly discussed, as the data were considered to be of insufficient robust quality for qualitative analysis. Level of evidence was based on the combination of study quality together with the total number of studies that measured the determinant. 'Strong evidence' for the price-modifying effect of a determinant was suggested if it was reported by three or more studies with an average 'fair quality' or higher and comparable associations were found. 'Moderate evidence' for the modifying effect on price or price variance of a determinant was suggested if it was reported by two 'fair quality' studies on average or higher t reported comparable associations. Pooled results with fewer studies, miscellaneous results or studies of poorer quality were considered 'poor' or 'conflicting', depending on these factors.

## Results

### Study selection

After duplication removal and title-abstract screening, 85 full-text articles were screened for eligibility. After full-text screening, 31 of these 85 articles were included in this systematic review. The reason for exclusion was often an irrelevant study outcome, such as drug efficacy, affordability, or study design, such as reviews discussing policy. A detailed process outline is visualized in the supplementary materials.

### Quality assessment

Twelve articles of good quality, fourteen articles of fair quality and five articles of poor quality were identified. The criteria that caused the most differentiation in article quality were adjustment for confounding factors, sample size



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justification/power description/variance and effect estimates. Table 1 presents a summary of the risk of bias assessment.

## Study characteristics

Of the 31 included studies, most had a cross-sectional design. Seven studies measured drug prices over a period of one to seventeen years, most often with more than one follow-up. 25 studies compared international prices (from here: prices) between countries, most often in high-income continents or countries such as Europe and/or the United States (US). Only the study by Van Harten et al. reported net drug prices. [5] 21 studies reported retail prices (of which three government prices, two wholesale retail prices and one out-of-pocket prices) and nine studies reported ex-factory prices (Table 1). Therefore, unless otherwise specified, drug prices indicate retail prices, not net drug prices, below. Table 1 displays study characteristics.

## Determinants

Table 1 describes all determinants measured per study. Table 2 depicts the best-evidence synthesis to which this section refers. Table 1 of the Supplementary Materials includes all determinants that were measured less than three times and were therefore not included in the best evidence synthesis. Below, the most common determinants of drug prices and drug price variance are described in further detail.

**National wealth**

Thirteen studies associated drug prices with national wealth, which was measured as a country's GDP or Gross National Income (GNI) per capita. Determinants in European countries were often studied because of comparable national wealth but fairly different drug pricing policies, allowing for more accurate evaluation of pricing policy efficacy. Three studies compared the most affluent European regions to other European countries, of which one included net drug prices. Seven studies compared the United States to other high-income countries. Six studies compared prices within a country, once in Austria, India, Kenya and three times in the US. These studies addressed differences between hospitals, pharmacies, subregions and states, respectively.

While in various countries drugs became cheaper or prices remained stable over time, prices tended to increase in the US compared to other western countries, in some instances up to 10% per year. [23-25]

Medication in Northern Europe was often more expensive compared to southern Europe, although this finding was inconsistent across studies and there was no consistent association between net cancer drug prices and GDP in high-income European countries. [5, 26-28] Conversely, after adjustment for PPP (purchasing power parity), higher GDP was associated with lower prices. [26-28]

Across a broad spectrum in high-income countries as well as high versus middle-income countries, an association between GDP and drug prices could not be established. [5, 11] Between middle-income and low-income countries, a moderately positive association between GDP and prices was found. After PPP adjustment, there was a direct relationship between GDP category and the frequency of lowest prices: across high, middle and low-income countries, high-income countries most often had the lowest prices. [29, 30] PPP-adjusted prices were especially high in Africa when compared to low- and middle-income countries. [31, 32] No clear association between drug prices and GDP could be established across low-income countries.

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### Market and drugs

Various studies assessed market influence in the form of a high proportion of generics or originators (nine studies) or patent expiry (three studies). Price-technical constructs such as discounts and mark-ups were common in five studies, of which the study by Van Harten et al. included net drug prices and four described retail prices. [5] Drug studies often focused on anti-cancer drugs (nine studies) or drugs for very specific patient groups such as hepatitis C, Alzheimer's disease or drugs for ophthalmic use. Occasionally, drugs were a random (representative) sample of widely used drugs, such as statins, analgesics and broad-spectrum antibiotics, most often including generics and originator drugs.

High market proportions of originator drugs led to high prices as well as price variance in all countries. In low and low-middle income countries, price variance is already significant. [33] In addition, a large originator price variance arises over time in the absence of price regulation. This association is not observed for generic substitutes. Drug prices in systems allowing for mark-ups tend to be higher than systems without. Strong evidence for price reduction was found in countries with a high generics proportion in the pharmaceutical market. Patent expiry was always accompanied by price reduction. Conflicting evidence suggests that a more independent status of pharmacies (compared to a chain) is associated with lower drug prices. [34, 35]

### Policies and government

Various studies assessed the association of drug prices with specific policy measures (3 studies for tendering, 3 for regulation scheme/indexation, 3 for central purchasing including one study reporting net drug prices). Of all price regulating policy measures, the strongest evidence exists for tendering. [27, 36, 37] Pricing regulation schemes and indexation schemes reduce price and price variance compared to unregulated market pricing. [23, 24, 37] Central purchasing by governmental agencies seems to lead to lower prices compared to semipublic or private purchasing. This effect, however, does not hold for net cancer drug prices across European countries with the exception of France. [5, 36, 37]

## Discussion

This study was conducted to systematically review (inter)national benchmark studies about the modifiable determinants of drug prices .

### Main findings

We found that literature on net drug prices is very scarce and that there is very limited insight in drug pricing mechanisms, indicative of an opaque price market. We are, therefore, not able to comment on determinants of net drug prices, which are the prices actually paid by the hospital or healthcare system.

Retail prices in an international and national context display a high degree of variance and not all variance can be explained by the measured determinants in this study. Despite this, we found that modifiable determinants that were associated with lower prices are: a higher generics market portion, discounts, tendering policies, central (governmental) purchasing and pricing regulation schemes. Modifiable determinants that were associated with higher prices are markets with higher originators market portion and systems in which mark-ups were common. In addition, the highest and most persistently increasing prices were seen in the United States, even compared to other high-income countries. A higher GDP was associated with higher prices in middle versus low-income countries, but not in

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high versus middle-income countries. After PPP adjustment, highest relative prices were consistently reported in low-income countries.

**Turning modifiable determinants into policies**

The drug market has been subject to dramatic inflation in size as well as price over the last decades, in which expensive drugs such as anticancer drugs stand out. [38] For policymakers of national governments, it is of paramount importance to know which alterations to the market and which government policies lead to lower, more affordable drug prices. Our findings about generics and originator market portions indicate that increasing the supply of generics is associated with price reductions, but this is often not the case as a stand-alone measure. For example, high-income countries with a relatively open market entry consistently experience higher initial prices for generics compared to countries with a more regulated generics market, such as Scandinavian countries with mandatory generic substitution. [36] In these countries, prices are half those in the Netherlands and Germany, which have a relatively open market. [39] Therefore, additional pricing regulation schemes might be necessary to actually achieve such a shift, and our results show various effective schemes. Direct price control by regulation on expensive drugs such as anticancer drugs likely has more influence than percental discounts. For example, indexation to average inflation of these drugs in the United Kingdom led to an annual inflation of only a quarter percent, compared to almost 9% in the unregulated market of the US over a period of approximately 10 years. [24] In any case, pricing policy in a predominantly originator market is less likely to lead to significant price reductions than generics-promoting policy. [40, 41]

Tendering led to lower prices and more specifically, lower generic-originator differences in various European countries. The risks of tendering include a (too) narrow focus on price and not quality. This may be of even higher importance in countries with limited quality control than in Europe and the US, where market approval is granted after a diligent selection process by the European Medicines Agency and Food and Drug Administration. [36] In addition, attention to tender frequency is important because prolonged supply exclusivity might outmarket other suppliers, leading to worsening oligopoly or even monopoly. Conversely, because of this exact reason, too high tender frequency reduces competitive pressure. [42] Tiered pricing agreements have been successful in reducing drugs prices in developing countries multifold and are relatively easy to implement, although, within the scope of our study, we have not been able to collect additional proof of this. [28]

**Regional and country-specific differences**

Due to the lack of significant price differences between middle and high-income countries, prices did not linearly increase with GDP throughout the entire GDP spectrum. In addition, the association of GDP and drug prices reversed after PPP adjustment.

On average, a 33% increase in price was found for every 50% increase in (unadjusted) wealth per capita for a representative set of World Health Organization essential drugs. [15] This observation is supported by the upsurge of pharmaceutical expenditure observed in steadily developing low- and middle-income countries. [43] Large price variance is noted by studies in low-income countries. This variance is largely explained by the relatively unorganized market and lack of pricing regulation. In contrast, developed countries generally have more organized price referencing systems and direct agreements with pharmaceutical companies, which means more expensive drugs can be purchased in large-scale volumes. [44] Although innovative drugs in low-income countries are gradually becoming

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more accessible, the introduction of expensive originator alternatives of essential drugs will likely result in limited health gain but significantly increased cost. [45]

A large portion of the observed price differences between high-income countries originates from the United States, likely because of the unregulated free-market approach. For example, the initial price listing of expensive (cancer) drugs in the UK is negotiated by the government, in contrast to the relatively open market entry in the US, leading to 42% lower prices in the UK. [24] Currently, countries in Europe have a tendency to dovetail their healthcare system design, leading to lower and more normalized price distributions across the continent, mostly due to external price referencing. [21] Indeed, consistent with our findings, this leads to further divergence from the United States. [46]

**Limitations**

The findings are to be interpreted with caution, because it is likely that large heterogeneity and the lack of (net) pricing data added significantly to the weighting in this best-evidence analyses and the outcomes of the synthesis. Nonetheless, a significant number of studies reported no association between the measured determinant(s) and drug price(s), indicating that not only positive findings were published. We recognize that the included studies only establish associations. For identifying causal associations, controlled trials are needed which are very hard to execute due to a lack of transparency and collaborative effort, as well as the difficulty of conducting an intervention *ceteris paribus*. Many studies reported large measures of uncertainty due to non-disclosure of data such as discounts, net prices paid, volumes sold, negotiations and arrangements with pharmaceutical companies. Therefore, many studies did not report an exact representation of the pharmaceutical market. Actual prices paid may have been more economical than the retail prices reported in the literature on which this review is based, but current literature is insufficient to give further insight in net drug price determinants. This exacerbates the challenge of reducing healthcare costs.

In addition, selection bias (medication selection) is likely significant as the most complete or readily available data is used, instead of drug price data that might be less available, such as price data on confidential pricing agreements. On the contrary, the total set of included studies comprises a large variety of drugs and countries, increasing external validity of these findings.

**Further research**

The lack of transparency and the high prevalence of non-disclosure agreements are likely two of the major drivers for price differences. In addition, we identified a gap in the representation of national studies, as only six national (within-country) analyses were identified. Therefore, we were unable solidly identify determinants of price in the national setting. Additional transparent, open-access research using net drug prices into price variance is needed to identify determinants of national and international price variance, which can be helpful to governments and regional institutes such as hospitals alike to improve their purchasing strategy and methods.

# Conclusion

Peer-reviewed literature on the determinants of net drug paid prices and corresponding pricing mechanisms is very scarce. This is remarkable given rising health care costs and in particular rising pharmaceutical costs, as well the growing societal discussion on mitigating healthcare expenditure. Retail drug prices display a high degree of variance, both within and between countries. This systematic review revealed empirical evidence of determinants retailing to a country's income level, its pharmaceutical market system and its policies and government. Policymakers may be



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particularly interested in the policy-related determinants of lower retail prices. Tendering, central purchasing and certain regulation schemes were consistently found to be negatively associated with retail prices. Various easily implementable governmental interventions and regulations, as well as modifiable market system-related characteristics, are consistently associated with lower retail prices and thus potential price reduction, such as increasing the generic portions of the drug market, governmental policies such as tendering, central purchasing and regulation schemes targeting annual price inflation, patent regulation and mark-ups. Drug prices in low-income countries are consistently higher after adjustment for PPP, indicating that lower-income countries still pay the highest relative retail prices. Uniform introduction of pricing regulation and publication of net drug prices will lead to more price transparency and a more balanced playing field of countries with a significant budget constrain. Such a situation allows for fair negotiations and ultimately a more sustainable healthcare sector. Most prominently, national and international net price benchmark studies are necessary to increase price transparency and to quantify the effectiveness of specific policy measures on net drug prices.

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# References


1. Keehan, S.P., et al., *National health expenditure projections, 2014-24: spending growth faster than recent trends.* Health Aff (Millwood), 2015. **34**(8): p. 1407-17.
2. Murthy, V.N.R. and A.A. Okunade, *Determinants of U.S. health expenditure: Evidence from autoregressive distributed lag (ARDL) approach to cointegration.* Economic Modelling, 2016. **59**: p. 67-73.
3. *The Global Use of Medicine in 2019 and Outlook to 2013: Forecasts and Areas to Watch.* IQVIA Institute for Human Data Science, 2019.
4. Hernandez, I., et al., *Changes in List Prices, Net Prices, and Discounts for Branded Drugs in the US, 2007-2018.* JAMA, 2020. **323**(9): p. 854-862.
5. van Harten, W.H., et al., *Actual costs of cancer drugs in 15 European countries.* Lancet Oncol, 2016. **17**(1): p. 18-20.
6. De Block, M., *The difficulty of comparing drug prices between countries.* Lancet Oncol, 2016. **17**(4): p. e125.
7. Dana, K.N., J.B. Hertig, and R.J. Weber, *Drug Pricing Transparency: The New Retail Revolution.* Hosp Pharm, 2017. **52**(2): p. 155-159.
8. Rintoul, A., et al., *Medicines with one seller and many buyers: strategies to increase the power of the payer.* BMJ, 2020. **369**: p. m1705.
9. Shrader, K.R., R.J. DeSalvo, and R.V. Evanson, *A comparison study of drug prices in segmented markets.* J Am Pharm Assoc, 1971. **11**(11): p. 598-602.
10. Vogler, S., K. Kilpatrick, and Z.U. Babar, *Analysis of Medicine Prices in New Zealand and 16 European Countries.* Value Health, 2015. **18**(4): p. 484-92.
11. Vogler, S., A. Vitry, and Z.U. Babar, *Cancer drugs in 16 European countries, Australia, and New Zealand: a cross-country price comparison study.* Lancet Oncol, 2016. **17**(1): p. 39-47.
12. Vogler, S., et al., *How Can Pricing and Reimbursement Policies Improve Affordable Access to Medicines? Lessons Learned from European Countries.* Appl Health Econ Health Policy, 2017. **15**(3): p. 307-321.
13. Ferrario, A., et al., *Strategies to achieve fairer prices for generic and biosimilar medicines.* BMJ, 2020. **368**: p. l5444.
14. Kanavos, P.G. and S. Vandoros, *Determinants of branded prescription medicine prices in OECD countries.* Health Econ Policy Law, 2011. **6**(3): p. 337-67.
15. Hanlon, M. and R. Zhang, *Price discrimination in essential medicines: evidence from International Drug Price Indicator Guide data.* Int Health, 2013. **5**(1): p. 58-63.
16. Russo, G. and B. McPake, *Medicine prices in urban Mozambique: a public health and economic study of pharmaceutical markets and price determinants in low-income settings.* Health Policy Plan, 2010. **25**(1): p. 70-84.
17. Yao, Y. and M. Tanaka, *Price offers of pharmaceutical procurement in China: evidence from Guangdong province.* Eur J Health Econ, 2016. **17**(5): p. 563-75.
18. Dylst, P. and S. Simoens, *Does the market share of generic medicines influence the price level?: a European analysis.* Pharmacoeconomics, 2011. **29**(10): p. 875-82.
19. Puig-Junoy, J., *Impact of European pharmaceutical price regulation on generic price competition: a review.* Pharmacoeconomics, 2010. **28**(8): p. 649-63.
20. Moher, D., et al., *Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.* BMJ, 2009. **339**: p. b2535.
21. *WHO Guideline on Country Pharmaceutical Pricing Policies* World Health Organization, 2015.
22. Health, N.I.o., *Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies.* Study Quality Assessment Tools, 2020.
23. Schlenker, M.B., G.E. Trope, and Y.M. Buys, *Comparison of United States and canadian glaucoma medication costs and price change from 2006 to 2013.* J Ophthalmol, 2015. **2015**: p. 547960.
24. Savage, P., et al., *Cancer Drugs: An International Comparison of Postlicensing Price Inflation.* J Oncol Pract, 2017. **13**(6): p. e538-e542.
25. Muzumdar, S., R. Wu, and H. Feng, *Temporal comparison of biologic medication prices between the United States and Australia.* J Dermatolog Treat, 2020: p. 1-2.
26. Leopold, C., et al., *Is Europe still heading to a common price level for on-patent medicines? An exploratory study among 15 Western European countries.* Health Policy, 2013. **112**(3): p. 209-16.
27. Wouters, O.J. and P.G. Kanavos, *A comparison of generic drug prices in seven European countries: a methodological analysis.* BMC Health Serv Res, 2017. **17**(1): p. 242.




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28. Iyengar, S., et al., *Prices, Costs, and Affordability of New Medicines for Hepatitis C in 30 Countries: An Economic Analysis.* PLoS Med, 2016. **13**(5): p. e1002032.
29. Salmasi, S., et al., *Pricing appraisal of anti-cancer drugs in the South East Asian, Western Pacific and East Mediterranean Region.* BMC Cancer, 2017. **17**(1): p. 903.
30. Suh, G.H., et al., *International price comparisons of Alzheimer's drugs: a way to close the affordability gap.* Int Psychogeriatr, 2009. **21**(6): p. 1116-26.
31. Cuomo, R.E., R.L. Seidman, and T.K. Mackey, *Country and regional variations in purchase prices for essential cancer medications.* BMC Cancer, 2017. **17**(1): p. 566.
32. Cameron, A., et al., *Medicine prices, availability, and affordability in 36 developing and middle-income countries: a secondary analysis.* Lancet, 2009. **373**(9659): p. 240-9.
33. Ongarora, D., et al., *Medicine Prices, Availability, and Affordability in Private Health Facilities in Low-Income Settlements in Nairobi County, Kenya.* Pharmacy (Basel), 2019. **7**(2).
34. Scott, A.T., P.E. Pecen, and A.G. Palestine, *Ophthalmic medication price variation across the United States: Anti-inflammatory medications.* Ther Adv Ophthalmol, 2019. **11**: p. 2515841419863638.
35. Theisen, K.M., et al., *Extreme Price Variation for Generic Benign Prostatic Hyperplasia Medications.* Urology, 2019. **124**: p. 223-228.
36. S., V., *How large are the differences between originator and generic prices? Analysis of five molecules in 16 European countries.* Economics, 2012. **DOI:10.7175/fe.v13i3S.311**.
37. Roughead, E.E., et al., *Pricing policies for generic medicines in Australia, New Zealand, the Republic of Korea and Singapore: patent expiry and influence on atorvastatin price.* WHO South East Asia J Public Health, 2018. **7**(2): p. 99-106.
38. Howard, D.H., et al., *Pricing in the Market for Anticancer Drugs.* J Econ Perspect, 2015. **29**(1): p. 139-62.
39. Simoens, S., *International comparison of generic medicine prices.* Curr Med Res Opin, 2007. **23**(11): p. 2647-54.
40. Sarpatwari, A., et al., *Competition and price among brand-name drugs in the same class: A systematic review of the evidence.* PLoS Med, 2019. **16**(7): p. e1002872.
41. Aalto-Setala, V., *The impact of generic substitution on price competition in Finland.* Eur J Health Econ, 2008. **9**(2): p. 185-91.
42. Maniadakis, N., et al., *Shaping Pharmaceutical Tenders for Effectiveness and Sustainability in Countries with Expanding Healthcare Coverage.* Appl Health Econ Health Policy, 2018. **16**(5): p. 591-607.
43. *World Medicines Situation - Medicines Expenditures.* World Health Organization, 2011.
44. Maniadakis, N., et al., *Comprehensive taxonomy and worldwide trends in pharmaceutical policies in relation to country income status.* BMC Health Serv Res, 2017. **17**(1): p. 371.
45. Zaidi, S., et al., *Access to essential medicines in Pakistan: policy and health systems research concerns.* PLoS One, 2013. **8**(5): p. e63515.
46. European Parliament, D.-g.f.i.p., *Differences in Costs of and Access to Pharmaceutical Products in the EU.* 2011.


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## Table 1: Study overview

| | Bulfone | Cameron | Clarke | Cuomo | Goldstein | Gong | Van Harten | Hill | Iyengar | Kolasani | Leopold | Mahlich | Manova | Muzumdar | Ongarora |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | 2009 | 2009 | 2010 | 2017 | 2017 | 2017 | 2016 | 2015 | 2016 | 2016 | 2013 | 2019 | 2018 | 2020 | 2019 |
| **Design** | Cross-sectional | Cross-sectional | **Longitudinal** | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional |
| **Setting** | International | International | International | International | International | National (US) and International | International | International | International | National | International | International | International | International | National |
| **Countries** | High-income (Australia and US) | Low and middle-income countries | High-income (Australia and UK) | Low and middle-income countries | Middle and high-income | High-income | High-income | High and middle-income | High, middle and low-income countries | Low-middle income (India) | High-income | High-income | High-income (Europe) | High-income (Australia and US) | Kenya |
| **Prices** | Retail prices | Government prices | Annual wholesale prices | Buyer prices | Retail and discounted prices | Wholesale, commercial, governmental health plan prices | Net drug prices | Lowest available price *and* target price | Ex-factory prices | Retail prices | Ex-factory price | Retail prices | Manufacturer and retail prices | Manufacturer prices | Retail price |
| **Drugs** | Generics | Essential generics and originator | Statins | Essential cancer | Patented cancer drugs | Ophthalmology | Cancer drugs (monoclonal antibodies) | TKIs with patent expiry < 5 yrs | Hepatitis C | Cancer drugs | Miscellaneous on-patent drugs | Abirateron | Rheumatoid arthritis biologicals and biosimilars | Etanercept, adalimumab, certolizumab, ustekinumab | Miscellaneous generic and originator drugs |
| **Study quality** | Fair | Good | Fair | Good | Good | Fair | Poor | Fair | Good | Poor | Good | Good | Good | Fair | Fair |
| **Association determinant and outcome** | **Transparency**: - (qualitative) **Discount**: - (qualitative) | **Mark-up: +** (high in generics) **GDP:** NS (low vs. middle)**, Geographic** Africa + **Government** tax on generics: + **Originator: +** (although smaller mark-ups) | **Life cycle:** fraction of generics -, patent expiry - | **GDP:** NS (low vs. middle) **Geographic:** Africa + **Life cycle**: NS for time passing, introduction of generics - | **GDP:** NS (high vs. middle, PPP-adjusted -) | **Governmental health plan: - Geographic:** US + **Originator: +** | **Governmental**: central purchasing: NS, France - **Geographic:** GDP NS (high-income) **Discounts:** - (Spain) | **Geographic:** US + (vs. Europe), India - (vs middle-income), high vs. middle-income - **Availability of generics**: - | **Government:** tiered pricing agreements - **GDP**: + (low vs. middle), NS (high vs. middle) **Geographic:** US + (unadjusted), Eastern Europe + (adjusted) | **Life cycle:** number of manufacturers NS, patent + (variance & price) | **Geographic:** West-/Northern Europe + (vs. Southern) Recession - **Life cycle:** NS over time | **Geographic:** NS | **Life cycle**: Patent expiry - | **Geographic**: US+ (vs. Australia), **Life cycle**: longitudinal: US+, Australia - | High variance in and between originator and generics |

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| | Roughead | Roughead | Salmasi | Savage | Schlenker | Scott | Simoens | Srivastava | Suh | Theisen | Vögler | Vögler | Vögler | Vögler | Vögler | Wouters |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | 2007 | 2018 | 2017 | 2017 | 2015 | 2019 | 2007 | 2014 | 2009 | 2019 | 2012 | 2013 | 2016 | 2015 | 2017 | 2017 |
| **Design** | **Cross-sectional** | **Longitudinal** | Cross-sectional | **Longitudinal** | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional |
| **Setting** | International | International | International | International | International | National | International | International | International | National | International | National | International | International | International | International |
| **Countries** | High-income (Australia and US) | High-income | High, middle and low-income | High-income (US and UK) | High-income | High-income (US) | High-income and India (middle) | Low-income | High, middle and low-income | United States | High-income | Austria (high-income) | High-income | High-income | High-income | High-income |
| **Prices** | Government price and ex-factory price | Retail prices | Retail prices | Retail prices | Retail prices | Retail price | Ex-factory price | Government price | Retail prices | Out-of-pocket price | Ex-factory prices | Retail prices | Ex-factory price | Ex-factory price | Ex-factory price | Ex-factory and retail price |
| **Drugs** | On-patent 'significant advancement' drugs | Atorvastatin (generic) | Originator anti-cancer drugs | High-volume anti-cancer drugs | Glaucoma medication | Ophthalmic | Generic medications in ambulatory care | Miscellaneous generic and originator drugs | Alzheimer's medications | Generic BPH medications | Miscellaneous generic and originator drugs | Miscellaneous generic and originator drugs | Originator cancer drugs | Miscellaneous generic and originator drugs | High-cost originator drugs | Miscellaneous generic drugs |
| **Study quality** | Good | Good | Fair | Poor | Poor | Good | Poor | Good | Fair | Good | Fair | Fair | Fair | Fair | Fair | Fair |
| **Association determinant and outcome** | **Government**: Government buying scheme - **Geographic**: US and Australia NS | **Policy:** Tendering -, regulation scheme -, patent expiry -, | **GDP:** + (high-income vs. middle and low), NS (high vs. high-middle income), - (low-income vs. high and middle income) | **Government**: price regulation scheme - **Geographic**: US + (vs. UK) **Life cycle**: US+ | **Government**: EPR-, price coupling to consumer price index **Geographic:** US + (vs. Canada), **Longitudinal price development**: US+, Canada -. | **Geography:** NS **Type of pharmacy** NS | **Geographic**: Mature generics market: -, India - **Free market**: + | **Mark-ups +, GDP** NS, **Life cycle**: originator variance ++ (vs. generic) | **GDP**: + (only unadjusted) | **Geographic:** high national variance**,** type of pharmacy: independency - (chain>wholesale>independent) | **Government:** No generic pricing policy, variance + Frequent tendering - Reference pricing - Free market, variance + **Geographic:** Scandinavian, variance + Size of generics market NS | **Discounts:** - (in-hospital, less expensive medicines) **Life cycle:** On-patent often no discount **Mark-ups**: + | **GDP**: + (high-income) **Life cycle**: *generics availability* - | **GDP**: + (high-income) **Life cycle**: *generics availability* - | **GDP:** + (high-income) | **Government**: *generics tendering* - **Geographic:** Northern Europe - (vs. Southern) |

**Abbreviations:**
BPH = benign prostate hypertrophy
GDP = gross domestic product
NS = non-significant outcome
PPP = purchasing power parity (*adjusted or unadjusted for PPP*)
US = United States

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**Table 2: Best-evidence synthesis**

| Category | Determinants | Studies | Effect | References | Level of evidence |
|---|---|---|---|---|---|
| Non-modifiable (Geographic) | **Regional/location** | | | | |
| | *GDP (high vs middle-income)* | 2 HQ, 3 FQ | NS, NS, +, NS, + | [28-30, 47, 48] | **Moderate for no association** |
| | *GDP (middle vs. low-income)* | 3 HQ, 2 FQ | NS, NS, **+, +, +** | [28-32] | **Moderate for positive association** |
| | *Europe (west/north vs. south/east)* | 2 HQ, 1 FQ | +, -, - | [26-28] | Conflicting evidence |
| | *United States (vs. other high-income)* | 2 HQ, 3 FQ, 1 LQ | NS, **+, +, +, +,+,+** | [23-25, 28, 48-50] | **Strong for positive association** |
| Modifiable | **Market characteristics** | | | | |
| | *High originator proportion* | 1 HQ, 1 FQ, 1 LQ | **+, +, +** | [31, 32, 49] | **Strong for positive association** |
| | *High generics proportion* | 4 FQ, 1 HQ, 1 LQ | **-, -, -,** NS**, -, -** | [10, 11, 36, 39, 48, 51] | **Strong for negative association** |
| | **Drug life cycle** | | | | |
| | *Time-longitudinal price development* | 2 HQ, 2 LQ | NS, NS, +, + | [23, 24, 26, 31] | Conflicting evidence |
| | *Patent expiry* | 2 HQ, 1 FQ | **-, -, -** | [37, 51, 52] | **Strong for negative association** |
| | **Discounts** | 2 FQ, 1 LQ | -, -, NS | [5, 53, 54] | **Moderate for negative association** |
| | **Mark-up** | 2 HQ, 1 FQ | **+, +, +** | [32, 53, 55] | **Strong for positive association** |
| Modifiable | **Policy** | | | | |
| | *Tendering* | 1 HQ, 2 FQ | **-, -, -** | [27, 36, 37] | **Strong for negative association** |
| | *Central purchaser* | 1 HQ, 1 FQ, 1 LQ | **-, -,** NS | [5, 36, 50] | **Moderate for negative association** |
| | *Pricing regulation (indexation) scheme* | 1 HQ, 2 LQ | **-, -, -** | [23, 24, 37] | **Moderate for negative association** |

**Abbreviations**

GDP: gross domestic product, HQ: high quality, FQ: fair quality, LQ: low quality, NS: not significant, + : positive effect, - : negative effect

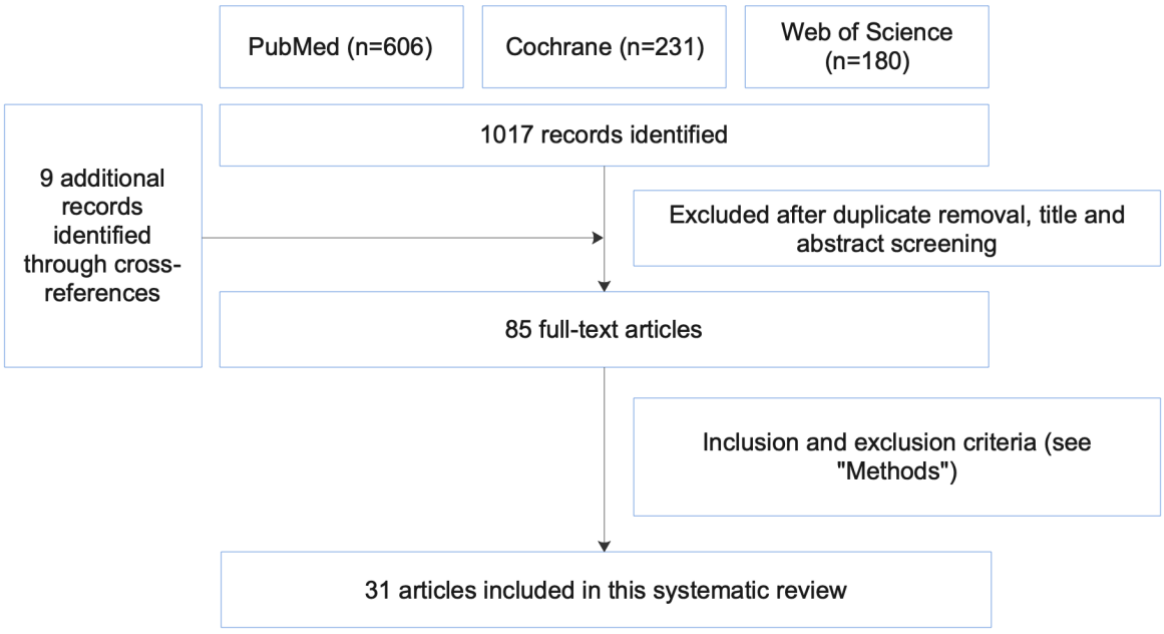
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# Supplementary materials

**eTable 1: PubMed Search Strategy**

(((((((("Pharmaceutical Preparations"[Mesh]) OR ((medicine[Title/Abstract] OR medicines[Title/Abstract] OR drug[Title/Abstract] OR drugs[Title/Abstract] OR pharmaceutical[Title/Abstract] OR pharmaceuticals[Title/Abstract] OR medication[Title/Abstract] OR medications[Title/Abstract])))) AND ((price[Title/Abstract] OR prices[Title/Abstract] OR pricing[Title/Abstract])))) AND (("Benchmarking"[Mesh]) OR ((benchmark[Title/Abstract] OR compare[Title/Abstract] OR comparison[Title/Abstract] OR comparisons[Title/Abstract])))))) NOT cost-effectiveness AND (full text[sb])

**eFigure 1: Study selection process**




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**eTable 2: Other determinants not in best-evidence synthesis (< 3 records)**

| Category | Determinant | Studies | Effect | References |
|---|---|---|---|---|
| **Geography** | *Africa (vs. low-income)* | 2 HQ | +, + | [31, 32] |
| | *France* | 1 LQ | + | [5] |
| | *India* | 2 FQ | -,- | [39, 48] |
| **(Pharmaceutical) market** | *Number of manufacturers* | 1 LQ | NS | [56] |
| | *Tiered pricing agreement* | 1 HQ | - | [28] |
| | *Independency of pharmacy* | 2 HQ | -, NS | [34, 37] |
| | *Recession* | 1 HQ | - | [26] |
| **Policy** | *Governmental health plan* | 1 FQ | - | [49] |
| | *EPR* | 1 LQ, 1 FQ | -, - | [23, 36] |
| | *Transparency* | 1 FQ | - | [54] |
| | *Free market* | 1 LQ | + | [39] |
| | *Generics tax* | 1 HQ | + | [32] |

HQ: high quality, FQ: fair quality, LQ: low quality, NS: not significant, + : positive effect, - : negative effect, EPR: external price referencing

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | 1 |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | 2 |
| **INTRODUCTION** | | | 3 |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | 3 |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | NA |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | 4 |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | 4 |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | 17 |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | 4 |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | 4 |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | 4 |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | 5 |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | 5 |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | 5 |

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | 5 |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | 5 |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | 5 |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | 6 |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | 14-15 |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | 6,7,14,15 |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | 6,7,14,15,16,18 |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | 9 |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | 16,18 |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | 7,8 |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | 9 |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | 9,10 |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | 10 |

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*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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# Determinants of Drug Prices: A Systematic Review of Comparison Studies



| | |
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# Determinants of Drug Prices: A Systematic Review of Comparison Studies

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**Keywords :** drug pricing, health economics, determinants, gross domestic product, regulation

# Abstract

**Objective**: In this systematic review on drug price comparison studies, we report on recent determinants of drug prices in a national and international context to facilitate the regulation of drug prices by purchasers and policymakers worldwide. Determinants of drug prices were divided into non-modifiable and modifiable and categorized as pertaining to a country's income level, pharmaceutical market system and its policies and government.

**Primary outcome**: determinants of drug prices or price variance.

**Design**: We systematically searched PubMed, EMBASE, Web of Science and Cochrane Library for peer-reviewed articles published after 2004 that reported an association of the primary outcome with one or more determinants. We performed a best-evidence synthesis of these associations for determinants that were covered in at least three studies.

**Results**: 31 publications were included, of which only one publication described net drug prices and 30 described retail drug prices. Five modifiable determinants were associated with lower retail prices: generic market portion, discounts, tendering policies, central (governmental) purchasing and pricing regulation schemes. The originators market portion and a system in which mark-ups are common were associated with higher retail prices. Retail prices were highest in the United States, even compared to other high-income countries. A positive association between national income level and drug retail prices could not be established among middle- and high-income countries. Retail prices were highest in low-income countries when adjusted for purchasing power parity.

**Conclusions**: Literature on determinants of net drug prices is extremely sparse. Various healthcare system interventions, market-specific and governmental regulations are consistently associated with lower retail prices. Some interventions are easily implementable in developing or middle-income countries, such as tendering, central purchasing and fixed pricing regulation schemes. Net drug price comparison studies are needed to overcome the lack of price transparency and to quantify the effectiveness of policy measures on net drug prices.

**300 words**


# Strengths and limitations of this study

- A systematic review on drug price comparison studies in peer-reviewed literature was conducted to assess all recently identified modifiable and non-modifiable determinants of drug prices.
- Only one published study conducted determinant analyses with net drug prices, while the rest included retail drug prices.
- The structured best-evidence synthesis is able to quantify heterogeneous evidence and summarizes the association between drug price (variance) and the reported determinants.
- This study design only allows for correlative analysis and cannot establish direct causal associations between determinants and price (variance).

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# Introduction

The global average share of healthcare costs in gross domestic product (GDP) increased from 4.6% of GDP in 1970 to 9.0% of GDP in 2016. [1] Several factors contributed to this rise, such as ageing populations, technological advances, the development of (expensive) new drugs and increased insurance coverage. Consequently, countries worldwide face the challenge of decreasing this healthcare cost burden. A significant driver of rising health care costs is pharmaceutical expenditure, so interventions to decrease drug costs through lower drug prices are warranted. Pharmaceutical expenditure is expected to reach 1.5 trillion dollars in 2023. [2, 3] Currently, a lack of transparency due to contract and pricing agreements between purchasers and manufacturers of drugs limits insight into net drug prices, because price comparisons are based on retail prices. Net drug prices are the prices actually paid by the purchasing healthcare institute, while retail prices exclude various individual pricing agreements such as rebates. [4] The effects of various policy measures are, therefore, unavailable to most purchasers and policymakers, which makes the drug price market opaque and actual comparisons of prices impossible. [5-8] The global drug price discussion is tremendous, but it remains unclear how drug prices are determined.

By comparing drug prices within and across different countries, determinants of drug prices can be identified. We make a distinction between determinants that are modifiable by policymakers and determinants that are not. Modifiable determinants could act as a target for potential interventions to decrease drug costs. Present drug policies are highly heterogeneous and have varying effects on drug prices. Therefore, identifying and harmonizing effective price management strategies may aid in accomplishing lower prices. However, international and national price comparison studies of net drug prices are scarce in the literature, which forms a bottleneck to compare drug prices. To our knowledge, the first publication that quantified drug price differences was performed in 1971 in the United States. [9] In this study, the net price paid by the United States (US) government was compared to the purchasing price of community pharmacies for the 50 most prescribed medications. Central governmental purchasing by the US resulted in a relative discount of nearly 50% over local pharmacies. Since then, research on drug prices (excluding price ratios) and, more specifically, net drug prices has been very limited, but in recent years some determinants of retail drug prices were identified. [5, 10-13] Most reported determinants are non-modifiable (e.g., geographical factors, GDP and product age). However, some are modifiable. In individual studies, systems allowing for mark-ups (e.g. processing costs or profit passed on to purchaser), degree of market competitiveness, the generics market proportion and degree of governmental intervention (reference pricing and education about efficacy of generics) have been found to influence drug prices in some countries. [14-19] A systematic analysis of modifiable drug price determinants that are published in drug price comparison studies is, however, missing.

Therefore, in this systematic review, we aim to identify determinants of drug prices and more specifically modifiable determinants that may serve as potential targets for drug price optimization. In addition, we give an overview of the recent literature in drug price comparisons and execute a best-evidence synthesis of all reported determinants of drug prices.

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# Methods

This systematic review was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. [20]

## Patient and public involvement

There were no patients involved in this study.

## Search strategy and information sources

The databases of PubMed, EMBASE, Web of Science and the Cochrane Library were searched until July 2020. We aimed to investigate which determinants influence drug prices by using comparison studies that measure at least one such determinant. All synonyms of the three topics of this study, *prices*, *drugs/medicines* and *benchmark*/*comparison*, were included in the search strategy - with the exclusion of *cost-effectiveness* - acknowledging the respective database formats (eTable 1 of the Supplementary Materials). All peer-reviewed articles that reported drug prices (excluding price ratios) in a benchmark or comparative study were included. We did not include primary grey literature documents, such as documents of non-governmental organizations, as our aim was to assess academic evidence of modifiable determinants, written by independent observers and published in peer-reviewed journals. Both national and international studies were included, as well as pharmacy or hospital comparisons, prospective, transverse and retrospective studies. Only articles published between 2004 until and including 2020 were included for two reasons. First, our aim is to investigate current determinants of drug prices. Second, articles on this topic published before 2004 were scarce and likely less relevant, as the pharmaceutical market underwent significant changes in the last fifteen years. [21] Articles were excluded if they only reported (a combination of) policies, efficacy, cost-effectiveness studies or studies examining general treatment costs (such as *diagnosis treatment combinations* [DBCs] in the Netherlands). We executed a two-way process of reference and citation checking of the included articles.

## Study eligibility and outcomes

The PICO (research format of population, intervention, control, outcome) consisted of drug or drug group (P), present determinant (I), non-present determinant (C), drug price or price variance (O). Therefore, all peer-reviewed studies that reported a measure of association between drug prices (meaning: not relative prices or price indices) or price variance (dependent variable) and one or more determinants (independent variable(s)) were included.

## Selection process

First, titles and abstracts (or if unavailable, full text) were independently screened by two researchers (JMJD and AdA). The full texts of potentially relevant articles were independently assessed for inclusion by two researchers (JMJD and AdA). Discrepancies between the researchers were resolved through discussion and consensus. If no consensus was reached, a third researcher (BvdB) was consulted.

## Data collection and extraction

A standardized extraction worksheet was used to collect study characteristics, methods and outcome information from all included articles by both AdA and JMJD. The extraction worksheet was ultimately cross-checked. The following

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items were collected: first author name, year of publication, manuscript title, country of study, study design, selection criteria (regarding included medications, institutes), (type of) price comparison/benchmark, studied determinants of drug prices and association between determinant and drug price.

### Risk of bias in individual studies and across studies

Quality assessment of the included studies was done with the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies of the National Institute of Health (NIH). [22] As the main purpose of this study is to give an overview of the field and to conduct a best-evidence synthesis, this instrument was only used to assess the quality and characteristics of the included studies and not to exclude additional studies. This specific assessment tool was used, because studies were mostly cross-sectional, sometimes observational and generally heterogeneous. As this tool is only meant to guide researchers in assessing such article's strengths and weaknesses, potential consequences of the identified risk of bias were interpreted by three authors (JMJD, AdA, BvdB) and consensus was reached based mainly on potential consequences on research results, as is also recommended by the NIH. In compliance with the assessment tool methods, studies were categorized in three quality levels: Good (all relevant qualifications met), Fair (1 or 2 not met) or Poor (3 or more not met).

Within the studies, selective reporting was assessed by carefully exploring the entire results section and supplementary material for non-reported associations.

### Summary measures and evidence synthesis

A qualitative best-evidence synthesis was performed to identify determinants of drug prices and variance in drug prices. Drug price variance reflects the potentiality of price differences of the same drug arising with comparable exposure to the analyzed determinant. Unfortunately, the heterogeneity between studies meant a quantitative analysis was not feasible. Determinants measured in every study were counted. If a determinant was studied three times or more, then it was analyzed and a best-evidence synthesis was conducted. If the determinant was studied once or twice, it was only shortly discussed, as the data were considered to be of insufficient robust quality for qualitative analysis. Level of evidence was based on the combination of study quality together with the total number of studies that measured the determinant. 'Strong evidence' for the price-modifying effect of a determinant was suggested if it was reported by three or more studies with an average 'fair quality' or higher and comparable associations were found. 'Moderate evidence' for the modifying effect on price or price variance of a determinant was suggested if it was reported by two 'fair quality' studies on average or higher t reported comparable associations. Pooled results with fewer studies, miscellaneous results or studies of poorer quality were considered 'poor' or 'conflicting', depending on these factors.

## Results

### Study selection

After duplication removal and title-abstract screening, 85 full-text articles were screened for eligibility. After full-text screening, 31 of these 85 articles were included in this systematic review. The reason for exclusion was often an irrelevant study outcome, such as drug efficacy, affordability, or study design, such as reviews discussing policy. A detailed process outline is visualized in the eFigure 1 of the Supplementary Materials.



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## Quality assessment

Ten articles of good quality, sixteen articles of fair quality and five articles of poor quality were identified. The criteria that caused the most differentiation in article quality were adjustment for confounding factors, sample size justification/power description/variance and effect estimates. Table 1 presents a summary of the risk of bias assessment.

## Study characteristics

Of the 31 included studies, most had a cross-sectional design. Seven studies measured drug prices over a period of one to seventeen years, most often with more than one follow-up. 25 studies compared international prices (from here: prices) between countries, most often in high-income continents or countries such as Europe and/or the US. Only the study by Van Harten et al. reported net drug prices. [5] 21 studies reported retail prices (of which three government prices, two wholesale retail prices and one out-of-pocket prices) and nine studies reported ex-factory prices (Table 1). Therefore, unless otherwise specified, drug prices indicate retail prices. Table 1 displays study characteristics.

## Determinants

Table 1 describes all determinants measured per study and categorizes the scope of the study in international versus national context. Table 2 depicts the best-evidence synthesis to which this section refers. eTable 2 of the Supplementary Materials includes all determinants that were measured less than three times and were therefore not included in the best evidence synthesis. Below, the most common determinants of drug prices and drug price variance are described in further detail.

**National wealth**

Thirteen studies associated drug prices with national wealth, which was measured as a country's GDP or Gross National Income (GNI) per capita. Determinants in European countries were often studied because of comparable national wealth but fairly different drug pricing policies, allowing for more accurate evaluation of pricing policy efficacy. Three studies compared the most affluent European regions to other European countries, of which one included net drug prices. Seven studies compared the US to other high-income countries. Six studies compared prices within a country, once in Austria, India, Kenya and three times in the US. These studies addressed differences between hospitals, pharmacies, subregions and states, respectively.

While in various countries drugs became cheaper or prices remained stable over time, prices tended to increase in the US compared to other western countries, in some instances up to 10% per year. [23-25]

Medication in Northern Europe was often more expensive compared to southern Europe, although this finding was inconsistent across studies and there was no consistent association between cancer drug prices and GDP in high-income European countries. [5, 26-28] Conversely, after adjustment for PPP (purchasing power parity), higher GDP was associated with lower prices. [26-28]

Across a broad spectrum in high-income countries as well as high versus middle-income countries, an association between GDP and drug prices could not be established. [5, 11] Between middle-income and low-income countries, a moderately positive association between GDP and prices was found. After PPP adjustment, there was a direct relationship between GDP category and the frequency of lowest prices: across high, middle and low-income countries, high-income countries most often had the lowest prices. [29, 30] PPP-adjusted prices were especially high in Africa

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when compared to low- and middle-income countries. [31, 32] No clear association between drug prices and GDP could be established across low-income countries.

**Market and drugs**

Various studies assessed market influence in the form of a high proportion of generics or originators (nine studies) or patent expiry (three studies). Price-technical constructs such as discounts and mark-ups were common in five studies, of which the study by Van Harten et al. included net drug prices and four described retail prices. [5] Drug studies often focused on anti-cancer drugs (nine studies) or drugs for very specific patient groups such as hepatitis C, Alzheimer's disease or drugs for ophthalmic use. Occasionally, drugs were a random (representative) sample of widely used drugs, such as statins, analgesics and broad-spectrum antibiotics, most often including generics and originator drugs.

High market proportions of originator drugs led to high prices as well as price variance in all countries. In low and low-middle income countries, price variance is already significant. [33] In addition, a large originator price variance arises over time in the absence of price regulation. This association is not observed for generic substitutes. Drug prices in systems allowing for mark-ups tend to be higher than systems without. Strong evidence for price reduction was found in countries with a high generics proportion in the pharmaceutical market. Patent expiry was always accompanied by price reduction. Conflicting evidence suggests that a more independent status of pharmacies (compared to a chain) is associated with lower drug prices. [34, 35]

**Policies and government**

Various studies assessed the association of drug prices with specific policy measures (3 studies for tendering, 3 for regulation scheme/indexation, 3 for central purchasing including one study reporting net drug prices). Of all price regulating policy measures, the strongest evidence exists for tendering. [27, 36, 37] Pricing regulation schemes and indexation schemes reduce price and price variance compared to unregulated market pricing. [23, 24, 37] Central purchasing by governmental agencies seems to lead to lower prices compared to semipublic or private purchasing. This effect, however, does not hold for net cancer drug prices across European countries with the exception of France. [5, 36, 37]

# Discussion

This study was conducted to systematically review drug price comparison studies about the modifiable determinants of drug prices.

**Main findings**

We found that literature on net drug prices is very scarce and that there is very limited insight in drug pricing mechanisms, which makes it impossible for purchasers and policymakers to transparently compare drug prices. These findings are indicative of an opaque price market. As the literature only delivers anecdotical evidence for differences in net drug prices, this study is unable to comment on determinants of net drug prices to accomplish insight in pricing mechanisms which can fuel the global drug price discussion. [5]

Retail prices in an international and national context display a high degree of variance and not all variance can be explained by the measured determinants in this study. Despite this, we found that modifiable determinants that were

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associated with lower prices are: a higher generics market portion, discounts, tendering policies, central (governmental) purchasing and pricing regulation schemes. Modifiable determinants that were associated with higher prices are markets with higher originators market portion and systems in which mark-ups were common. In addition, the highest and most persistently increasing prices were seen in the US, even compared to other high-income countries. A higher GDP was associated with higher prices in middle versus low-income countries, but not in high versus middle-income countries. After PPP adjustment, highest relative prices were consistently reported in low-income countries.

**Turning modifiable determinants into policies**

The drug market has been subject to dramatic inflation in size as well as price over the last decades, in which expensive drugs such as anticancer drugs stand out. [38] For policymakers of national governments, it is of paramount importance to know which alterations to the market and which government policies lead to lower, more affordable drug prices. Our findings about generics and originator market portions indicate that increasing the supply of generics is associated with price reductions, but this is often not the case as a stand-alone measure. For example, high-income countries with a relatively open market entry consistently experience higher initial prices for generics compared to countries with a more regulated generics market, such as Scandinavian countries with mandatory generic substitution. [36] In these countries, prices are half those in the Netherlands and Germany, which have a relatively open market. [39] Therefore, additional pricing regulation schemes might be necessary to actually achieve such a shift, and our results show various effective schemes. Direct price control by regulation on expensive drugs such as anticancer drugs likely has more influence than percental discounts. For example, indexation to average inflation of these drugs in the United Kingdom led to an annual inflation of only a quarter percent, compared to almost 9% in the unregulated market of the US over a period of approximately 10 years. [24] In any case, pricing policy in a predominantly originator market is less likely to lead to significant price reductions than generics-promoting policy. [40, 41]

Tendering led to lower prices and more specifically, lower generic-originator differences in various European countries. The risks of tendering include a (too) narrow focus on price and not quality. This may be of even higher importance in countries with limited quality control than in Europe and the US, where market approval is granted after a diligent selection process by the European Medicines Agency and Food and Drug Administration. [36] In addition, attention to tender frequency is important because prolonged supply exclusivity might outmarket other suppliers, leading to worsening oligopoly or even monopoly. Conversely, because of this exact reason, too high tender frequency reduces competitive pressure. [42] Tiered pricing agreements have been successful in reducing drugs prices in developing countries multifold and are relatively easy to implement, although, within the scope of our study, we have not been able to collect additional proof of this. [28] All in all, no intervention should be evaluated on its own; new interventions need to be weighted in the total ecosystem of interventions. Each intervention has different consequences based on the timing and sequence of implementation. Moreover, many interdependencies exist between existing measures and new interventions. [43] One intervention is likely to have different outcomes in different healthcare systems. An earlier extensive review on determinants of pharmaceutical expenditure has also shed light on these dynamic properties of market interventions and governmental regulation on drug prices (as well as on investments in research and development). [44]

Therefore, we recommend future policy effectivity studies to include a careful assessment of (external) validity of their results; an extensive description of the healthcare system, governmental regularization and market properties.

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**Regional and country-specific differences**

Due to the lack of significant price differences between middle and high-income countries, prices did not linearly increase with GDP throughout the entire GDP spectrum. In addition, the association of GDP and drug prices reversed after PPP adjustment.

On average, a 33% increase in price was found for every 50% increase in (unadjusted) wealth per capita for a representative set of World Health Organization (WHO) essential drugs. [15] This observation is supported by the upsurge of pharmaceutical expenditure observed in steadily developing low- and middle-income countries. [45] Large price variance is noted by studies in low-income countries. This variance is largely explained by the relatively unorganized market and lack of pricing regulation. In contrast, developed countries generally have more organized price referencing systems and direct agreements with pharmaceutical companies, which means more expensive drugs can be purchased in large-scale volumes. [46] Although innovative drugs in low-income countries are gradually becoming more accessible, the introduction of expensive originator alternatives of essential drugs will likely result in limited health gain but significantly increased cost. [47]

A large portion of the observed price differences between high-income countries originates from the US, likely because of the unregulated free-market approach. For example, the initial price listing of expensive (cancer) drugs in the UK is negotiated by the government, in contrast to the relatively open market entry in the US, leading to 42% lower prices in the UK. [24] Currently, countries in Europe have a tendency to dovetail their healthcare system design, leading to lower and more normalized price distributions across the continent, mostly due to external price referencing. [21] Indeed, consistent with our findings, this leads to further divergence from the US. [48]

**Limitations**

The findings are to be interpreted with caution, because it is likely that large heterogeneity and the lack of net pricing data added significantly to the weighting in this best-evidence analyses and the outcomes of the synthesis. Nonetheless, a significant number of studies reported no association between the measured determinant(s) and drug price(s), indicating that not only positive findings were published.

The lack of net pricing is likely partly due to the fact that we excluded primary grey literature documents from our main research corpus. Various institutions, most prominently the WHO, are increasing their efforts in assessing pharmaceutical expenditure and its determinants as shown by a recent overview of grey literature [49] Research institutes have set-up various inspiring collaborations with governmental and non-governmental organizations (NGO) using national databanks (such as Austria's Pharma Price Information, or PPI) or World Health Organization (WHO) data, which are rich and often available at request. [32, 50, 51] In addition, WHO now promotes and collects various drug price databases and occasionally publishes in peer-reviewed journals. [32, 52] Ideally, these studies give rise to further collaboration and NGO-supervised observational studies in developing countries, allowing academics to assess effectiveness of interventions on net drug prices.

We recognize that the included studies only establish associations. For identifying causal associations, controlled trials are needed which are very hard to execute due to a lack of transparency and collaborative effort, as well as the difficulty of conducting an intervention *ceteris paribus*. Many studies reported large measures of uncertainty due to non-disclosure of data such as discounts, net prices paid, volumes sold, negotiations and arrangements with pharmaceutical companies. Therefore, many studies did not report an exact representation of the pharmaceutical



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market. Net prices may have been more economical than the retail prices reported in the literature on which this review is based, but current literature is insufficient to give further insight in net drug price determinants. This exacerbates the challenge of reducing healthcare costs, as real price comparisons are can never be made in this state of affairs.

In addition, selection bias (medication selection) is likely significant as the most complete or readily available data is used, instead of drug price data that might be less available, such as price data on confidential pricing agreements. On the contrary, the total set of included studies comprises a large variety of drugs and countries, increasing external validity of these findings.

**Further research**

The lack of transparency and the high prevalence of non-disclosure agreements are likely two of the major drivers for drug price differences. We call for an elimination of non-disclosure agreements to better understand underlying drug price determinants and mechanisms. In addition, we identified a gap in the representation of national studies, as only six national (within-country) analyses were identified. Therefore, we were unable solidly identify determinants of price in the national setting.

Additional transparent, open-access research into price variance using net drug prices is needed to identify determinants of national and international price variance, which can be helpful to governments and regional institutes such as hospitals alike to improve their purchasing strategy and methods. Uniform introduction of pricing regulation and publication of net drug prices may ultimately lead to more price transparency and a more balanced playing field of countries with a significant budget constrain. Such a situation allows for fair negotiations and ultimately a more sustainable and transparent healthcare sector.

# Conclusion

Peer-reviewed literature on the determinants of net drug paid prices and corresponding pricing mechanisms is very scarce. This is remarkable given rising health care costs and in particular rising pharmaceutical costs, as well the growing societal discussion on mitigating healthcare expenditure. Retail drug prices display a high degree of variance, both within and between countries. This systematic review revealed empirical evidence of determinants retailing to a country's income level, its pharmaceutical market system and its policies and government. Policymakers may be particularly interested in the policy-related determinants of lower retail prices. Drug prices in low-income countries are consistently higher after adjustment for PPP, indicating that lower-income countries still pay the highest relative retail prices. Various easily implementable governmental interventions and regulations, as well as modifiable market system-related characteristics, are consistently associated with lower retail prices and thus potential price reduction, such as increasing the generic portions of the drug market, governmental policies such as tendering, central purchasing and regulation schemes targeting annual price inflation, patent regulation and mark-ups, which might aid developing countries in closing the gap of purchasing power-adjusted prices with high-income countries in the future. Most prominently, national and international net price studies are necessary to quantify the effectiveness of these policy measures on net drug prices. We call for openness of net drug price data, as this is our key to change the healthcare cost burden.

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## Contributor statement

All authors contributed to the conception and design of the research. JMJD and AdA analyzed the data and drafted the first manuscript version. BvdB and MVH provided in multiple sessions conceptual and detailed feedback for every version of the manuscript.

## Competing interests

The authors have no conflicts of interest to declare.

## Data sharing statement

All data relevant to this study are included in the article or available as supplementary material.

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# References


1. Keehan, S.P., et al., *National health expenditure projections, 2014-24: spending growth faster than recent trends.* Health Aff (Millwood), 2015. **34**(8): p. 1407-17.
2. Murthy, V.N.R. and A.A. Okunade, *Determinants of U.S. health expenditure: Evidence from autoregressive distributed lag (ARDL) approach to cointegration.* Economic Modelling, 2016. **59**: p. 67-73.
3. *The Global Use of Medicine in 2019 and Outlook to 2013: Forecasts and Areas to Watch.* IQVIA Institute for Human Data Science, 2019.
4. Hernandez, I., et al., *Changes in List Prices, Net Prices, and Discounts for Branded Drugs in the US, 2007-2018.* JAMA, 2020. **323**(9): p. 854-862.
5. van Harten, W.H., et al., *Actual costs of cancer drugs in 15 European countries.* Lancet Oncol, 2016. **17**(1): p. 18-20.
6. De Block, M., *The difficulty of comparing drug prices between countries.* Lancet Oncol, 2016. **17**(4): p. e125.
7. Dana, K.N., J.B. Hertig, and R.J. Weber, *Drug Pricing Transparency: The New Retail Revolution.* Hosp Pharm, 2017. **52**(2): p. 155-159.
8. Rintoul, A., et al., *Medicines with one seller and many buyers: strategies to increase the power of the payer.* BMJ, 2020. **369**: p. m1705.
9. Shrader, K.R., R.J. DeSalvo, and R.V. Evanson, *A comparison study of drug prices in segmented markets.* J Am Pharm Assoc, 1971. **11**(11): p. 598-602.
10. Vogler, S., K. Kilpatrick, and Z.U. Babar, *Analysis of Medicine Prices in New Zealand and 16 European Countries.* Value Health, 2015. **18**(4): p. 484-92.
11. Vogler, S., A. Vitry, and Z.U. Babar, *Cancer drugs in 16 European countries, Australia, and New Zealand: a cross-country price comparison study.* Lancet Oncol, 2016. **17**(1): p. 39-47.
12. Vogler, S., et al., *How Can Pricing and Reimbursement Policies Improve Affordable Access to Medicines? Lessons Learned from European Countries.* Appl Health Econ Health Policy, 2017. **15**(3): p. 307-321.
13. Ferrario, A., et al., *Strategies to achieve fairer prices for generic and biosimilar medicines.* BMJ, 2020. **368**: p. l5444.
14. Kanavos, P.G. and S. Vandoros, *Determinants of branded prescription medicine prices in OECD countries.* Health Econ Policy Law, 2011. **6**(3): p. 337-67.
15. Hanlon, M. and R. Zhang, *Price discrimination in essential medicines: evidence from International Drug Price Indicator Guide data.* Int Health, 2013. **5**(1): p. 58-63.
16. Russo, G. and B. McPake, *Medicine prices in urban Mozambique: a public health and economic study of pharmaceutical markets and price determinants in low-income settings.* Health Policy Plan, 2010. **25**(1): p. 70-84.
17. Yao, Y. and M. Tanaka, *Price offers of pharmaceutical procurement in China: evidence from Guangdong province.* Eur J Health Econ, 2016. **17**(5): p. 563-75.
18. Dylst, P. and S. Simoens, *Does the market share of generic medicines influence the price level?: a European analysis.* Pharmacoeconomics, 2011. **29**(10): p. 875-82.
19. Puig-Junoy, J., *Impact of European pharmaceutical price regulation on generic price competition: a review.* Pharmacoeconomics, 2010. **28**(8): p. 649-63.
20. Moher, D., et al., *Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.* BMJ, 2009. **339**: p. b2535.
21. *WHO Guideline on Country Pharmaceutical Pricing Policies* World Health Organization, 2015.
22. Health, N.I.o., *Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies.* Study Quality Assessment Tools, 2020.
23. Schlenker, M.B., G.E. Trope, and Y.M. Buys, *Comparison of United States and canadian glaucoma medication costs and price change from 2006 to 2013.* J Ophthalmol, 2015. **2015**: p. 547960.
24. Savage, P., et al., *Cancer Drugs: An International Comparison of Postlicensing Price Inflation.* J Oncol Pract, 2017. **13**(6): p. e538-e542.
25. Muzumdar, S., R. Wu, and H. Feng, *Temporal comparison of biologic medication prices between the United States and Australia.* J Dermatolog Treat, 2020: p. 1-2.
26. Leopold, C., et al., *Is Europe still heading to a common price level for on-patent medicines? An exploratory study among 15 Western European countries.* Health Policy, 2013. **112**(3): p. 209-16.
27. Wouters, O.J. and P.G. Kanavos, *A comparison of generic drug prices in seven European countries: a methodological analysis.* BMC Health Serv Res, 2017. **17**(1): p. 242.




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28. Iyengar, S., et al., *Prices, Costs, and Affordability of New Medicines for Hepatitis C in 30 Countries: An Economic Analysis.* PLoS Med, 2016. **13**(5): p. e1002032.
29. Salmasi, S., et al., *Pricing appraisal of anti-cancer drugs in the South East Asian, Western Pacific and East Mediterranean Region.* BMC Cancer, 2017. **17**(1): p. 903.
30. Suh, G.H., et al., *International price comparisons of Alzheimer's drugs: a way to close the affordability gap.* Int Psychogeriatr, 2009. **21**(6): p. 1116-26.
31. Cuomo, R.E., R.L. Seidman, and T.K. Mackey, *Country and regional variations in purchase prices for essential cancer medications.* BMC Cancer, 2017. **17**(1): p. 566.
32. Cameron, A., et al., *Medicine prices, availability, and affordability in 36 developing and middle-income countries: a secondary analysis.* Lancet, 2009. **373**(9659): p. 240-9.
33. Ongarora, D., et al., *Medicine Prices, Availability, and Affordability in Private Health Facilities in Low-Income Settlements in Nairobi County, Kenya.* Pharmacy (Basel), 2019. **7**(2).
34. Scott, A.T., P.E. Pecen, and A.G. Palestine, *Ophthalmic medication price variation across the United States: Anti-inflammatory medications.* Ther Adv Ophthalmol, 2019. **11**: p. 2515841419863638.
35. Theisen, K.M., et al., *Extreme Price Variation for Generic Benign Prostatic Hyperplasia Medications.* Urology, 2019. **124**: p. 223-228.
36. S., V., *How large are the differences between originator and generic prices? Analysis of five molecules in 16 European countries.* Economics, 2012. **DOI:10.7175/fe.v13i3S.311**.
37. Roughead, E.E., et al., *Pricing policies for generic medicines in Australia, New Zealand, the Republic of Korea and Singapore: patent expiry and influence on atorvastatin price.* WHO South East Asia J Public Health, 2018. **7**(2): p. 99-106.
38. Howard, D.H., et al., *Pricing in the Market for Anticancer Drugs.* J Econ Perspect, 2015. **29**(1): p. 139-62.
39. Simoens, S., *International comparison of generic medicine prices.* Curr Med Res Opin, 2007. **23**(11): p. 2647-54.
40. Sarpatwari, A., et al., *Competition and price among brand-name drugs in the same class: A systematic review of the evidence.* PLoS Med, 2019. **16**(7): p. e1002872.
41. Aalto-Setala, V., *The impact of generic substitution on price competition in Finland.* Eur J Health Econ, 2008. **9**(2): p. 185-91.
42. Maniadakis, N., et al., *Shaping Pharmaceutical Tenders for Effectiveness and Sustainability in Countries with Expanding Healthcare Coverage.* Appl Health Econ Health Policy, 2018. **16**(5): p. 591-607.
43. Sood N, d.V.H., Gutierrez Italo, Lakdawalla D, Goldman DP, *The Effect Of Regulation On Pharmaceutical Revenues: Experience In Nineteen Countries.* Health Affairs, 2008. **27**(Supplement 1): p. w125-w137.
44. Lakdawalla, D., *Economics of the Pharmaceutical Industry.* Journal of Economic Literature 2018, 56(2), 397–449, 2018.
45. *World Medicines Situation - Medicines Expenditures.* World Health Organization, 2011.
46. Maniadakis, N., et al., *Comprehensive taxonomy and worldwide trends in pharmaceutical policies in relation to country income status.* BMC Health Serv Res, 2017. **17**(1): p. 371.
47. Zaidi, S., et al., *Access to essential medicines in Pakistan: policy and health systems research concerns.* PLoS One, 2013. **8**(5): p. e63515.
48. European Parliament, D.-g.f.i.p., *Differences in Costs of and Access to Pharmaceutical Products in the EU.* 2011.
49. Borges Dos Santos, M.A., et al., *Factors influencing pharmaceutical pricing - a scoping review of academic literature in health science.* J Pharm Policy Pract, 2019. **12**: p. 24.
50. Vogler, S., N. Zimmermann, and Z.U. Babar, *Price comparison of high-cost originator medicines in European countries.* Expert Rev Pharmacoecon Outcomes Res, 2017. **17**(2): p. 221-230.
51. Vogler, S., P. Schneider, and N. Zimmermann, *Evolution of Average European Medicine Prices: Implications for the Methodology of External Price Referencing.* Pharmacoecon Open, 2019. **3**(3): p. 303-309.
52. WHO, *World Health Organization: Medicine price information sources.* https://www.who.int/medicines/areas/access/med_price_info_sources/en/, last accessed February 3rd, 2021.
53. Goldstein, D.A., et al., *A global comparison of the cost of patented cancer drugs in relation to global differences in wealth.* Oncotarget, 2017. **8**(42): p. 71548-71555.
54. Hill, A., et al., *Target prices for mass production of tyrosine kinase inhibitors for global cancer treatment.* BMJ Open, 2016. **6**(1): p. e009586.


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55. Gong, D., et al., *Comparison of United States and International Ophthalmic Drug Pricing.* Ophthalmology, 2019. **126**(10): p. 1358-1365.
56. Roughead, E.E., R. Lopert, and L.N. Sansom, *Prices for innovative pharmaceutical products that provide health gain: a comparison between Australia and the United States.* Value Health, 2007. **10**(6): p. 514-20.
57. Clarke, P.M. and E.M. Fitzgerald, *Expiry of patent protection on statins: effects on pharmaceutical expenditure in Australia.* Med J Aust, 2010. **192**(11): p. 633-6.
58. Manova, M., et al., *Comparative Price Analysis of Biological Products for Treatment of Rheumatoid Arthritis.* Front Pharmacol, 2018. **9**: p. 1070.
59. Vogler, S., et al., *The role of discounts and loss leaders in medicine procurement in Austrian hospitals - a primary survey of official and actual medicine prices.* Cost Eff Resour Alloc, 2013. **11**(1): p. 15.
60. Bulfone, L., *High prices for generics in Australia - more competition might help.* Aust Health Rev, 2009. **33**(2): p. 200-14.
61. Srivastava, D. and A. McGuire, *Analysis of prices paid by low-income countries - how price sensitive is government demand for medicines?* BMC Public Health, 2014. **14**: p. 767.
62. Kolasani, B.P., D.C. Malathi, and R.R. Ponnaluri, *Variation of Cost among Anti-cancer Drugs Available in Indian Market.* J Clin Diagn Res, 2016. **10**(11): p. FC17-FC20.

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## Table 1: Study overview

| | Bulfone | Cameron | Clarke | Cuomo | Goldstein | Van Harten | Hill | Iyengar | Leopold | Mahlich | Manova | Muzumdar | Roughead | Roughead | Salr |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | 2009 | 2009 | 2010 | 2017 | 2017 | 2016 | 2015 | 2016 | 2013 | 2019 | 2018 | 2020 | 2007 | 2018 | 201 |
| **Design** | Cross-sectional | Cross-sectional | **Longitudinal** | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | **Cross-sectional** | **Longitudinal** | Cros |
| **Setting** | International | International | International | International | International | International | International | International | International | International | International | International | International | International | Inte |
| **Countries** | High-income (Australia and US) | Low and middle-income countries | High-income (Australia and UK) | Low and middle-income countries | Middle and high-income | High-income | High and middle-income | High, middle and low-income countries | High-income | High-income | High-income (Europe) | High-income (Australia and US) | High-income (Australia and US) | High-income | High low- |
| **Prices** | Retail prices | Government prices | Annual wholesale prices | Buyer prices | Retail and discounted prices | Net drug prices | Lowest available price *and* target price | Ex-factory prices | Ex-factory price | Retail prices | Manufacturer and retail prices | Manufacturer prices | Government price and ex-factory price | Retail prices | Reta |
| **Drugs** | Generics | Essential generics and originator | Statins | Essential cancer | Patented cancer drugs | Cancer drugs (monoclonal antibodies) | TKIs with patent expiry < 5 yrs | Hepatitis C | Miscellaneous on-patent drugs | Abirateron | Rheumatoid arthritis biologicals and biosimilars | Etanercept, adalimumab, certolizumab, ustekinumab | On-patent 'significant advancement' drugs | Atorvastatin (generic) | Orig canc |
| **Study quality** | Fair | Good | Fair | Good | Good | Poor | Fair | Good | Good | Good | Good | Fair | Good | Good | Fair |
| **Association determinant and outcome** | **Transparency**: - (qualitative) **Discount**: - (qualitative) | **Mark-up:** + (high in generics) **GDP:** NS (low vs. middle), **Geographic** Africa + **Government** tax on generics: + **Originator:** + (although smaller mark-ups) | **Life cycle:** fraction of generics -, patent expiry - | **GDP:** NS (low vs. middle) **Geographic:** Africa + **Life cycle**: NS for time passing, introduction of generics - | **GDP:** NS (high vs. middle, PPP-adjusted -) | **Governmental**: central purchasing: NS, France - **Geographic:** GDP NS (high-income) **Discounts:** - (Spain) | **Geographic**: US + (vs. Europe), India - (vs middle-income), high vs. middle-income - **Availability of generics**: - | **Government:** tiered pricing agreements - **GDP**: + (low vs. middle), NS (high vs. middle) **Geographic:** US + (unadjusted), Eastern Europe + (adjusted) | **Geographic:** West-/Northern Europe + (vs. Southern) Recession - **Life cycle:** NS over time | **Geographic:** NS | **Life cycle**: Patent expiry - | **Geographic**: US+ (vs. Australia), **Life cycle**: longitudinal: US+, Australia - | **Government**: Government buying scheme - **Geographic**: US and Australia NS | **Policy:** Tendering -, regulation scheme -, patent expiry -, | **GD** inco mid NS high inco inco and inco |

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| | Savage | Schlenker | Simoens | Srivastava | Suh | Vögler | Vögler | Vögler | Vögler | Wouters | Gong | Kolasani | Ongarora | Scott | Theisen | Vögler |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | 2017 | 2015 | 2007 | 2014 | 2009 | 2012 | 2016 | 2015 | 2017 | 2017 | 2017 | 2016 | 2019 | 2019 | 2019 | 2013 |
| **Design** | **Longitudinal** | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional |
| **Setting** | International | International | International | International | International | International | International | International | International | International | National (US) and International | National | National | National | National | National |
| **Countries** | High-income (US and UK) | High-income | High-income and India (middle) | Low-income | High, middle and low-income | High-income | High-income | High-income | High-income | High-income | High-income | Low-middle income (India) | Kenya | High-income (US) | United States | Austria (high-income) |
| **Prices** | Retail prices | Retail prices | Ex-factory price | Government price | Retail prices | Ex-factory prices | Ex-factory price | Ex-factory price | Ex-factory price | Ex-factory and retail price | Wholesale, commercial, governmental health plan prices | Retail prices | Retail price | Out-of-pocket price | Out-of-pocket price | Retail prices |
| **Drugs** | High-volume anti-cancer drugs | Glaucoma medication | Generic medications in ambulatory care | Miscellaneous generic and originator drugs | Alzheimer's medications | Miscellaneous generic and originator drugs | Originator cancer drugs | Miscellaneous generic and originator drugs | High-cost originator drugs | Miscellaneous generic drugs | Ophthalmology | Cancer drugs | Miscellaneous generic and originator drugs | Ophthalmic | Generic BPH medications | Miscellaneous generic and originator drugs |
| **Study quality** | Poor | Poor | Poor | Good | Fair | Fair | Fair | Fair | Fair | Fair | Fair | Poor | Fair | Fair | Fair | Fair |
| **Association determinant and outcome** | **Government**: price regulation scheme - **Geographic**: US + (vs. UK) **Life cycle**: US+ | **Government**: EPR-, price coupling to consumer price index **Geographic:** US + (vs. Canada), **Longitudinal price development**: US+, Canada -. | **Geographic**: Mature generics market: -, India - **Free market**: + | **Mark-ups +, GDP** NS, **Life cycle**: originator variance ++ (vs. generic) | **GDP**: + (only unadjusted) | **Government:** No generic pricing policy, variance + Frequent tendering - Reference pricing - Free market, variance + **Geographic:** Scandinavian, variance + Size of generics market NS | **GDP**: + (high-income) **Life cycle**: *generics availability* - | **GDP**: + (high-income) **Life cycle**: *generics availability* - | **GDP:** + (high-income) | **Government**: *generics tendering* - **Geographic:** Northern Europe - (vs. Southern) | **Governmental health plan: - Geographic:** US + **Originator: +** | **Life cycle:** number of manufacturers NS, patent + (variance & price) | High variance in and between originator and generics | **Geography:** NS **Type of pharmacy** NS | **Geographic:** high national variance**,** type of pharmacy: independency - (chain>wholesale>independent) | **Discounts:** - (in-hospital, less expensive medicines) **Life cycle:** On-patent often no discount **Mark-ups**: + |

*Price category listed as mentioned or as described in the article.*
BPH = benign prostate hypertrophy
GDP = gross domestic product
NS = non-significant outcome
PPP = purchasing power parity (*adjusted or unadjusted for PPP*)
US = United States

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**Table 2: Best-evidence synthesis**

| Category | Determinants | Studies | Effect | References | Level of evidence |
|---|---|---|---|---|---|
| Non-modifiable (Geographic) | **Regional/location** | | | | |
| | *GDP (high vs middle-income)* | 2 HQ, 3 FQ | NS, NS, +, NS, + | [28-30, 53, 54] | **Moderate for no association** |
| | *GDP (middle vs. low-income)* | 3 HQ, 2 FQ | NS, NS, **+, +,** + | [28-32] | **Moderate for positive association** |
| | *Europe (west/north vs. south/east)* | 2 HQ, 1 FQ | +, **-, -** | [26-28] | Conflicting evidence |
| | *United States (vs. other high-income)* | 2 HQ, 3 FQ, 1 LQ | NS, **+, +, +, +,+,+** | [23-25, 28, 54-56] | **Strong for positive association** |
| Modifiable | **Market characteristics** | | | | |
| | *High originator proportion* | 1 HQ, 1 FQ, 1 LQ | **+, +,** + | [31, 32, 55] | **Strong for positive association** |
| | *High generics proportion* | 4 FQ, 1 HQ, 1 LQ | **-, -, -,** NS**, -, -** | [10, 11, 36, 39, 54, 57] | **Strong for negative association** |
| | **Drug life cycle** | | | | |
| | *Time-longitudinal price development* | 2 HQ, 2 LQ | NS, NS, +, + | [23, 24, 26, 31] | Conflicting evidence |
| | *Patent expiry* | 2 HQ, 1 FQ | **-, -, -** | [37, 57, 58] | **Strong for negative association** |
| | **Discounts** | 2 FQ, 1 LQ | **-, -,** NS | [5, 59, 60] | **Moderate for negative association** |
| | **Mark-up** | 2 HQ, 1 FQ | **+, +,** + | [32, 59, 61] | **Strong for positive association** |
| Modifiable | **Policy** | | | | |
| | *Tendering* | 1 HQ, 2 FQ | **-, -, -** | [27, 36, 37] | **Strong for negative association** |
| | *Central purchaser* | 1 HQ, 1 FQ, 1 LQ | **-, -,** NS | [5, 36, 56] | **Moderate for negative association** |
| | *Pricing regulation (indexation) scheme* | 1 HQ, 2 LQ | **-, -, -** | [23, 24, 37] | **Moderate for negative association** |

**Abbreviations**

GDP: gross domestic product, HQ: high quality, FQ: fair quality, LQ: low quality, NS: not significant, + : positive effect, - : negative effect

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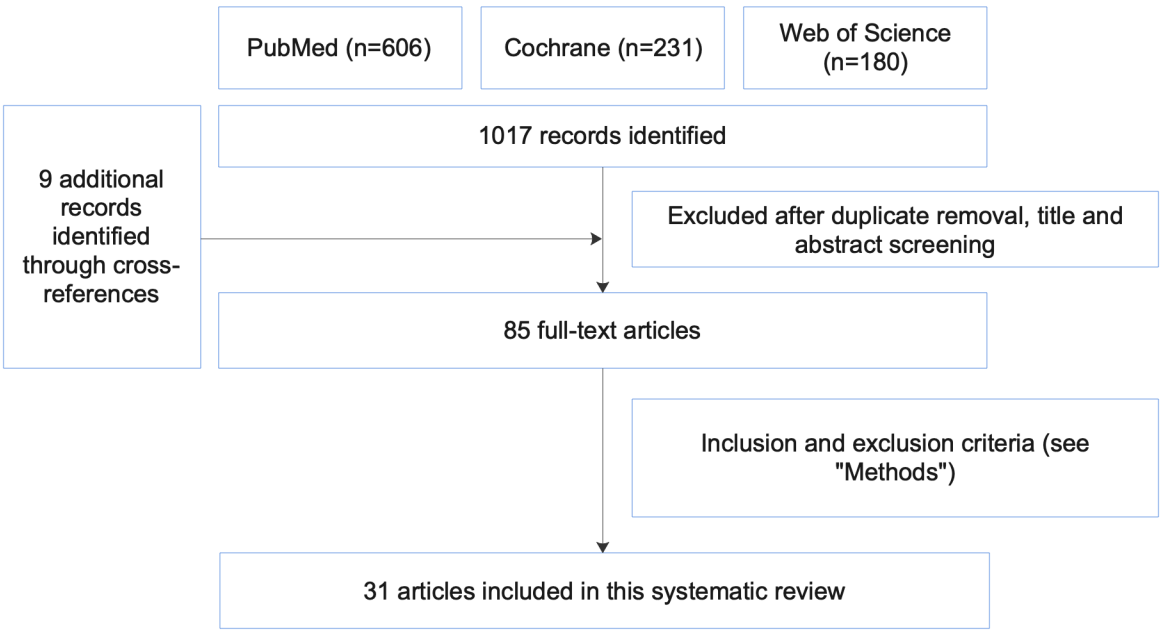
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# Supplementary materials

**eTable 1: PubMed Search Strategy**

(((((((("Pharmaceutical Preparations"[Mesh]) OR ((medicine[Title/Abstract] OR medicines[Title/Abstract] OR drug[Title/Abstract] OR drugs[Title/Abstract] OR pharmaceutical[Title/Abstract] OR pharmaceuticals[Title/Abstract] OR medication[Title/Abstract] OR medications[Title/Abstract]))) AND ((price[Title/Abstract] OR prices[Title/Abstract] OR pricing[Title/Abstract]))) AND (("Benchmarking"[Mesh]) OR ((benchmark[Title/Abstract] OR compare[Title/Abstract] OR comparison[Title/Abstract] OR comparisons[Title/Abstract])))) NOT cost-effectiveness AND (full text[sb])

**eFigure 1: Study selection process**




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**eTable 2: Other determinants not in best-evidence synthesis (< 3 records)**

Supplementary

| Category | Determinant | Studies | Effect | References |
|---|---|---|---|---|
| **Geography** | *Africa (vs. low-income)* | 2 HQ | +, + | [31, 32] |
| | *France* | 1 LQ | + | [5] |
| | *India* | 2 FQ | -,- | [39, 54] |
| **(Pharmaceutical) market** | *Number of manufacturers* | 1 LQ | NS | [62] |
| | *Tiered pricing agreement* | 1 HQ | - | [28] |
| | *Independency of pharmacy* | 2 FQ | -, NS | [34, 35] |
| | *Recession* | 1 HQ | - | [26] |
| **Policy** | *Governmental health plan* | 1 FQ | - | [55] |
| | *EPR* | 1 LQ, 1 FQ | -, - | [23, 36] |
| | *Transparency* | 1 FQ | - | [60] |
| | *Free market* | 1 LQ | + | [39] |
| | *Generics tax* | 1 HQ | + | [32] |

HQ: high quality, FQ: fair quality, LQ: low quality, NS: not significant, + : positive effect, - : negative effect, EPR: external price referencing

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | **1** |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | **2** |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | **3** |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | **3** |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | **-** |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | **4** |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | **4** |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | **4-+17** |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | **4** |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | **4-5** |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | **4-5** |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | **5** |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | **5** |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | **5** |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | **5** |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | **5** |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | **5-6** |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | **6** |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | **9-10** |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | **6-7-15-16** |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | **6-7** |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | **9-10** |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | **17** |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | **7-8-9** |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | **9-10** |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | **10** |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | **11** |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

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# Determinants of Drug Prices: A Systematic Review of Comparison Studies

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# Determinants of Drug Prices: A Systematic Review of Comparison Studies

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**Keywords :** drug pricing, health economics, determinants, gross domestic product, regulation

# Abstract

**Objective**: In this systematic review on drug price comparison studies, we report on recent determinants of drug prices in a national and international context to facilitate regulation of drug prices by purchasers and policymakers worldwide. Determinants of drug prices were divided into non-modifiable and modifiable and categorized as pertaining to a country's income level, pharmaceutical market system and its policies and government.

**Primary outcome**: determinants of drug prices or price variance.

**Design**: We systematically searched PubMed, EMBASE, Web of Science and Cochrane Library for peer-reviewed articles published between 2004 and July 22, 2020 that reported an association of the primary outcome with one or more determinants. We performed a best-evidence synthesis of these associations for determinants covered in at least three studies.

**Results**: 31 publications were included. Only one publication described net drug prices and 30 described retail drug prices. Five modifiable determinants were associated with lower retail prices: generic market portion, discounts, tendering policies, central (governmental) purchasing and pricing regulation schemes. The originators market portion and a system in which mark-ups are common were associated with higher retail prices. Retail prices were highest in the United States, even compared to other high-income countries. A positive association between national income level and drug retail prices could not be established among middle- and high-income countries. Retail prices were highest in low-income countries when adjusted for purchasing power parity.

**Conclusions**: Literature on determinants of net drug prices is extremely sparse. Various healthcare system interventions, market-specific and governmental regulations are consistently associated with lower retail prices. Some interventions are easily implementable in developing or middle-income countries, such as tendering, central purchasing and fixed pricing regulation schemes. Net drug price comparison studies are needed to overcome the lack of price transparency and to quantify the effectiveness of policy measures on net drug prices.

**299 words**


## Strengths and limitations of this study

- A systematic review on drug price comparison studies in peer-reviewed literature was conducted to assess all recently identified modifiable and non-modifiable determinants of drug prices.
- Only one published study conducted determinant analyses with net drug prices, while the rest included retail drug prices.
- The structured best-evidence synthesis is able to quantify heterogeneous evidence and summarizes the association between drug price (variance) and the reported determinants.
- This study design only allows for correlative analysis and cannot establish direct causal associations between determinants and price (variance).

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# Introduction

The global average share of healthcare costs in gross domestic product (GDP) increased from 4.6% of GDP in 1970 to 9.0% of GDP in 2016. [1] Several factors contributed to this rise, such as ageing populations, technological advances, the development of (expensive) new drugs and increased insurance coverage. Consequently, countries worldwide face the challenge of decreasing this healthcare cost burden. A significant driver of rising health care costs is pharmaceutical expenditure, so interventions to decrease drug costs through lower drug prices are warranted. Pharmaceutical expenditure is expected to reach 1.5 trillion dollars in 2023. [2, 3] Currently, a lack of transparency due to contract and pricing agreements between purchasers and manufacturers of drugs limits insight into net drug prices, because price comparisons are based on retail prices. Net drug prices are the prices actually paid by the purchasing healthcare institute, while retail prices exclude various individual pricing agreements such as rebates. [4] The effects of various policy measures are, therefore, unavailable to most purchasers and policymakers, which makes the drug price market opaque and actual comparisons of prices impossible. [5-8] The global drug price discussion is tremendous, but it remains unclear how drug prices are determined.

By comparing drug prices within and across different countries, determinants of drug prices can be identified. We make a distinction between determinants that are modifiable by policymakers and determinants that are not. Modifiable determinants could act as a target for potential interventions to decrease drug costs. Present drug policies are highly heterogeneous and have varying effects on drug prices. Therefore, identifying and harmonizing effective price management strategies may aid in accomplishing lower prices. However, international and national price comparison studies of net drug prices are scarce in the literature, which forms a bottleneck to compare drug prices. To our knowledge, the first publication that quantified drug price differences was performed in 1971 in the United States. [9] In this study, the net price paid by the United States (US) government was compared to the purchasing price of community pharmacies for the 50 most prescribed medications. Central governmental purchasing by the US resulted in a relative discount of nearly 50% over local pharmacies. Since then, research on drug prices (excluding price ratios) and, more specifically, net drug prices has been very limited, but in recent years some determinants of retail drug prices were identified. [5, 10-13] Most reported determinants are non-modifiable (e.g., geographical factors, GDP and product age). However, some are modifiable. In individual studies, systems allowing for mark-ups (e.g. processing costs or profit passed on to purchaser), degree of market competitiveness, the generics market proportion and degree of governmental intervention (reference pricing and education about efficacy of generics) have been found to influence drug prices in some countries. [14-19] A systematic analysis of modifiable drug price determinants that are published in drug price comparison studies is, however, missing.

Therefore, in this systematic review, we aim to identify determinants of drug prices and more specifically modifiable determinants that may serve as potential targets for drug price optimization. In addition, we give an overview of the recent literature in drug price comparisons and execute a best-evidence synthesis of all reported determinants of drug prices.

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# Methods

This systematic review was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) statement. [20]

## Patient and public involvement

There were no patients involved in this study.

## Search strategy and information sources

The databases of PubMed, EMBASE, Web of Science and the Cochrane Library were searched until July 22, 2020. We aimed to investigate which determinants influence drug prices by using comparison studies that measure at least one such determinant. All synonyms of the three topics of this study, *prices*, *drugs/medicines* and *benchmark*/*comparison*, were included in the search strategy - with the exclusion of *cost-effectiveness* - acknowledging the respective database formats (eTable 1 of the Supplementary Materials). All peer-reviewed articles that reported drug prices (excluding price ratios) in a benchmark or comparative study were included. We did not include primary grey literature documents, such as documents of non-governmental organizations, as our aim was to assess academic evidence of modifiable determinants, written by independent observers and published in peer-reviewed journals. Both national and international studies were included, as well as pharmacy or hospital comparisons, prospective, transverse and retrospective studies. Only articles published between 2004 until and including 2020 were included for two reasons. First, our aim is to investigate current determinants of drug prices. Second, articles on this topic published before 2004 were scarce and likely less relevant, as the pharmaceutical market underwent significant changes in the last fifteen years. [21] Articles were excluded if they only reported (a combination of) policies, efficacy, cost-effectiveness studies or studies examining general treatment costs (such as *diagnosis treatment combinations* [DBCs] in the Netherlands). We executed a two-way process of reference and citation checking of the included articles.

## Study eligibility and outcomes

The PICO (research format of population, intervention, control, outcome) consisted of drug or drug group (P), present determinant (I), non-present determinant (C), drug price or price variance (O). Therefore, all peer-reviewed studies that reported a measure of association between drug prices (meaning: not relative prices or price indices) or price variance (dependent variable) and one or more determinants (independent variable(s)) were included.

## Selection process

First, titles and abstracts (or if unavailable, full text) were independently screened by two researchers (JMJD and AdA). The full texts of potentially relevant articles were independently assessed for inclusion by two researchers (JMJD and AdA). Discrepancies between the researchers were resolved through discussion and consensus. If no consensus was reached, a third researcher (BvdB) was consulted.

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## Data collection and extraction

A standardized extraction worksheet was used to collect study characteristics, methods and outcome information from all included articles by both AdA and JMJD. The extraction worksheet was ultimately cross-checked. The following items were collected: first author name, year of publication, manuscript title, country of study, study design, selection criteria (regarding included medications, institutes), (type of) price comparison/benchmark, studied determinants of drug prices and association between determinant and drug price.

## Risk of bias in individual studies and across studies

Quality assessment of the included studies was done with the Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies of the National Institute of Health (NIH). [22] As the main purpose of this study is to give an overview of the field and to conduct a best-evidence synthesis, this instrument was only used to assess the quality and characteristics of the included studies and not to exclude additional studies. This specific assessment tool was used, because studies were mostly cross-sectional, sometimes observational and generally heterogeneous. As this tool is only meant to guide researchers in assessing such article's strengths and weaknesses, potential consequences of the identified risk of bias were interpreted by three authors (JMJD, AdA, BvdB) and consensus was reached based mainly on potential consequences on research results, as is also recommended by the NIH. In compliance with the assessment tool methods, studies were categorized in three quality levels: Good (all relevant qualifications met), Fair (1 or 2 not met) or Poor (3 or more not met).

Within the studies, selective reporting was assessed by carefully exploring the entire results section and supplementary material for non-reported associations.

## Summary measures and evidence synthesis

A qualitative best-evidence synthesis was performed to identify determinants of drug prices and variance in drug prices. Drug price variance reflects the potentiality of price differences of the same drug arising with comparable exposure to the analyzed determinant. Unfortunately, the heterogeneity between studies meant a quantitative analysis was not feasible. Determinants measured in every study were counted. If a determinant was studied three times or more, then it was analyzed and a best-evidence synthesis was conducted. If the determinant was studied once or twice, it was only shortly discussed, as the data were considered to be of insufficient robust quality for qualitative analysis. Level of evidence was based on the combination of study quality together with the total number of studies that measured the determinant. 'Strong evidence' for the price-modifying effect of a determinant was suggested if it was reported by three or more studies with an average 'fair quality' or higher and comparable associations were found. 'Moderate evidence' for the modifying effect on price or price variance of a determinant was suggested if it was reported by two 'fair quality' studies on average or higher t reported comparable associations. Pooled results with fewer studies, miscellaneous results or studies of poorer quality were considered 'poor' or 'conflicting', depending on these factors.

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# Results

## Study selection

After duplication removal and title-abstract screening, 85 full-text articles were screened for eligibility. After full-text screening, 31 of these 85 articles were included in this systematic review. The reason for exclusion was often an irrelevant study outcome, such as drug efficacy, affordability, or study design, such as reviews discussing policy. A detailed process outline is visualized in the eFigure 1 of the Supplementary Materials.

## Quality assessment

Ten articles of good quality, sixteen articles of fair quality and five articles of poor quality were identified. The criteria that caused the most differentiation in article quality were adjustment for confounding factors, sample size justification/power description/variance and effect estimates. Table 1 presents a summary of the risk of bias assessment.

## Study characteristics

Of the 31 included studies, most had a cross-sectional design. Seven studies measured drug prices over a period of one to seventeen years, most often with more than one follow-up. 25 studies compared international prices (from here: prices) between countries, most often in high-income continents or countries such as Europe and/or the US. Only the study by Van Harten et al. reported net drug prices. [5] 21 studies reported retail prices (of which three government prices, two wholesale retail prices and one out-of-pocket prices) and nine studies reported ex-factory prices (Table 1). Therefore, unless otherwise specified, drug prices indicate retail prices. Table 1 displays study characteristics.

## Determinants

Table 1 describes all determinants measured per study and categorizes the scope of the study in international versus national context. Table 2 depicts the best-evidence synthesis to which this section refers. eTable 2 of the Supplementary Materials includes all determinants that were measured less than three times and were therefore not included in the best evidence synthesis. Below, the most common determinants of drug prices and drug price variance are described in further detail.

### National wealth

Thirteen studies associated drug prices with national wealth, which was measured as a country's GDP or Gross National Income (GNI) per capita. Determinants in European countries were often studied because of comparable national wealth but fairly different drug pricing policies, allowing for more accurate evaluation of pricing policy efficacy. Three studies compared the most affluent European regions to other European countries, of which one included net drug prices. Seven studies compared the US to other high-income countries. Six studies compared prices within a country, once in Austria, India, Kenya and three times in the US. These studies addressed differences between hospitals, pharmacies, subregions and states, respectively.

While in various countries drugs became cheaper or prices remained stable over time, prices tended to increase in the US compared to other western countries, in some instances up to 10% per year. [23-25]

Medication in Northern Europe was often more expensive compared to southern Europe, although this finding was inconsistent across studies and there was no consistent association between cancer drug prices and GDP in high-



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income European countries. [5, 26-28] Conversely, after adjustment for PPP (purchasing power parity), higher GDP was associated with lower prices. [26-28]

Across a broad spectrum in high-income countries as well as high versus middle-income countries, an association between GDP and drug prices could not be established. [5, 11] Between middle-income and low-income countries, a moderately positive association between GDP and prices was found. After PPP adjustment, there was a direct relationship between GDP category and the frequency of lowest prices: across high, middle and low-income countries, high-income countries most often had the lowest prices. [29, 30] PPP-adjusted prices were especially high in Africa when compared to low- and middle-income countries. [31, 32] No clear association between drug prices and GDP could be established across low-income countries.

**Market and drugs**

Various studies assessed market influence in the form of a high proportion of generics or originators (nine studies) or patent expiry (three studies). Price-technical constructs such as discounts and mark-ups were common in five studies, of which the study by Van Harten et al. included net drug prices and four described retail prices. [5] Drug studies often focused on anti-cancer drugs (nine studies) or drugs for very specific patient groups such as hepatitis C, Alzheimer's disease or drugs for ophthalmic use. Occasionally, drugs were a random (representative) sample of widely used drugs, such as statins, analgesics and broad-spectrum antibiotics, most often including generics and originator drugs.

High market proportions of originator drugs led to high prices as well as price variance in all countries. In low and low-middle income countries, price variance is already significant. [33] In addition, a large originator price variance arises over time in the absence of price regulation. This association is not observed for generic substitutes. Drug prices in systems allowing for mark-ups tend to be higher than systems without. Strong evidence for price reduction was found in countries with a high generics proportion in the pharmaceutical market. Patent expiry was always accompanied by price reduction. Conflicting evidence suggests that a more independent status of pharmacies (compared to a chain) is associated with lower drug prices. [34, 35]

**Policies and government**

Various studies assessed the association of drug prices with specific policy measures (3 studies for tendering, 3 for regulation scheme/indexation, 3 for central purchasing including one study reporting net drug prices). Of all price regulating policy measures, the strongest evidence exists for tendering. [27, 36, 37] Pricing regulation schemes and indexation schemes reduce price and price variance compared to unregulated market pricing. [23, 24, 37] Central purchasing by governmental agencies seems to lead to lower prices compared to semipublic or private purchasing. This effect, however, does not hold for net cancer drug prices across European countries with the exception of France. [5, 36, 37]

# Discussion

This study was conducted to systematically review drug price comparison studies about the modifiable determinants of drug prices.

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**Main findings**

We found that literature on net drug prices is very scarce and that there is very limited insight in drug pricing mechanisms, which makes it impossible for purchasers and policymakers to transparently compare drug prices. These findings are indicative of an opaque price market. As the literature only delivers anecdotical evidence for differences in net drug prices, this study is unable to comment on determinants of net drug prices to accomplish insight in pricing mechanisms which can fuel the global drug price discussion. [5]

Retail prices in an international and national context display a high degree of variance and not all variance can be explained by the measured determinants in this study. Despite this, we found that modifiable determinants that were associated with lower prices are: a higher generics market portion, discounts, tendering policies, central (governmental) purchasing and pricing regulation schemes. Modifiable determinants that were associated with higher prices are markets with higher originators market portion and systems in which mark-ups were common. In addition, the highest and most persistently increasing prices were seen in the US, even compared to other high-income countries. A higher GDP was associated with higher prices in middle versus low-income countries, but not in high versus middle-income countries. After PPP adjustment, highest relative prices were consistently reported in low-income countries.

**Turning modifiable determinants into policies**

The drug market has been subject to dramatic inflation in size as well as price over the last decades, in which expensive drugs such as anticancer drugs stand out. [38] For policymakers of national governments, it is of paramount importance to know which alterations to the market and which government policies lead to lower, more affordable drug prices. Our findings about generics and originator market portions indicate that increasing the supply of generics is associated with price reductions, but this is often not the case as a stand-alone measure. For example, high-income countries with a relatively open market entry consistently experience higher initial prices for generics compared to countries with a more regulated generics market, such as Scandinavian countries with mandatory generic substitution. [36] In these countries, prices are half those in the Netherlands and Germany, which have a relatively open market. [39] Therefore, additional pricing regulation schemes might be necessary to actually achieve such a shift, and our results show various effective schemes. Direct price control by regulation on expensive drugs such as anticancer drugs likely has more influence than percental discounts. For example, indexation to average inflation of these drugs in the United Kingdom led to an annual inflation of only a quarter percent, compared to almost 9% in the unregulated market of the US over a period of approximately 10 years. [24] In any case, pricing policy in a predominantly originator market is less likely to lead to significant price reductions than generics-promoting policy. [40, 41]

Tendering led to lower prices and more specifically, lower generic-originator differences in various European countries. The risks of tendering include a (too) narrow focus on price and not quality. This may be of even higher importance in countries with limited quality control than in Europe and the US, where market approval is granted after a diligent selection process by the European Medicines Agency and Food and Drug Administration. [36] In addition, attention to tender frequency is important because prolonged supply exclusivity might outmarket other suppliers, leading to worsening oligopoly or even monopoly. Conversely, because of this exact reason, too high tender frequency reduces competitive pressure. [42] Tiered pricing agreements have been successful in reducing drugs prices in developing countries multifold and are relatively easy to implement, although, within the scope of our study, we have not been able to collect additional proof of this. [28] All in all, no intervention should be evaluated on its own; new



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interventions need to be weighted in the total ecosystem of interventions. Each intervention has different consequences based on the timing and sequence of implementation. Moreover, many interdependencies exist between existing measures and new interventions. [43] One intervention is likely to have different outcomes in different healthcare systems. An earlier extensive review on determinants of pharmaceutical expenditure has also shed light on these dynamic properties of market interventions and governmental regulation on drug prices (as well as on investments in research and development). [44]

Therefore, we recommend future policy effectivity studies to include a careful assessment of (external) validity of their results; an extensive description of the healthcare system, governmental regularization and market properties.

**Regional and country-specific differences**

Due to the lack of significant price differences between middle and high-income countries, prices did not linearly increase with GDP throughout the entire GDP spectrum. In addition, the association of GDP and drug prices reversed after PPP adjustment.

On average, a 33% increase in price was found for every 50% increase in (unadjusted) wealth per capita for a representative set of World Health Organization (WHO) essential drugs. [15] This observation is supported by the upsurge of pharmaceutical expenditure observed in steadily developing low- and middle-income countries. [45] Large price variance is noted by studies in low-income countries. This variance is largely explained by the relatively unorganized market and lack of pricing regulation. In contrast, developed countries generally have more organized price referencing systems and direct agreements with pharmaceutical companies, which means more expensive drugs can be purchased in large-scale volumes. [46] Although innovative drugs in low-income countries are gradually becoming more accessible, the introduction of expensive originator alternatives of essential drugs will likely result in limited health gain but significantly increased cost. [47]

A large portion of the observed price differences between high-income countries originates from the US, likely because of the unregulated free-market approach. For example, the initial price listing of expensive (cancer) drugs in the UK is negotiated by the government, in contrast to the relatively open market entry in the US, leading to 42% lower prices in the UK. [24] Currently, countries in Europe have a tendency to dovetail their healthcare system design, leading to lower and more normalized price distributions across the continent, mostly due to external price referencing. [21] Indeed, consistent with our findings, this leads to further divergence from the US. [48]

**Limitations**

The findings are to be interpreted with caution, because it is likely that large heterogeneity and the lack of net pricing data added significantly to the weighting in this best-evidence analyses and the outcomes of the synthesis. Nonetheless, a significant number of studies reported no association between the measured determinant(s) and drug price(s), indicating that not only positive findings were published.

The lack of net pricing is likely partly due to the fact that we excluded primary grey literature documents from our main research corpus. Various institutions, most prominently the WHO, are increasing their efforts in assessing pharmaceutical expenditure and its determinants as shown by a recent overview of grey literature [49] Research institutes have set-up various inspiring collaborations with governmental and non-governmental organizations (NGO) using national databanks (such as Austria's Pharma Price Information, or PPI) or World Health Organization (WHO) data, which are rich and often available at request. [32, 50, 51] In addition, WHO now promotes and collects various



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drug price databases and occasionally publishes in peer-reviewed journals. [32, 52] Ideally, these studies give rise to further collaboration and NGO-supervised observational studies in developing countries, allowing academics to assess effectiveness of interventions on net drug prices.

We recognize that the included studies only establish associations. For identifying causal associations, controlled trials are needed which are very hard to execute due to a lack of transparency and collaborative effort, as well as the difficulty of conducting an intervention *ceteris paribus*. Many studies reported large measures of uncertainty due to non-disclosure of data such as discounts, net prices paid, volumes sold, negotiations and arrangements with pharmaceutical companies. Therefore, many studies did not report an exact representation of the pharmaceutical market. Net prices may have been more economical than the retail prices reported in the literature on which this review is based, but current literature is insufficient to give further insight in net drug price determinants. This exacerbates the challenge of reducing healthcare costs, as real price comparisons are can never be made in this state of affairs.

In addition, selection bias (medication selection) is likely significant as the most complete or readily available data is used, instead of drug price data that might be less available, such as price data on confidential pricing agreements. On the contrary, the total set of included studies comprises a large variety of drugs and countries, increasing external validity of these findings.

**Further research**

The lack of transparency and the high prevalence of non-disclosure agreements are likely two of the major drivers for drug price differences. We call for an elimination of non-disclosure agreements to better understand underlying drug price determinants and mechanisms. In addition, we identified a gap in the representation of national studies, as only six national (within-country) analyses were identified. Therefore, we were unable solidly identify determinants of price in the national setting.

Additional transparent, open-access research into price variance using net drug prices is needed to identify determinants of national and international price variance, which can be helpful to governments and regional institutes such as hospitals alike to improve their purchasing strategy and methods. Uniform introduction of pricing regulation and publication of net drug prices may ultimately lead to more price transparency and a more balanced playing field of countries with a significant budget constrain. Such a situation allows for fair negotiations and ultimately a more sustainable and transparent healthcare sector.

# Conclusion

Peer-reviewed literature on the determinants of net drug paid prices and corresponding pricing mechanisms is very scarce. This is remarkable given rising health care costs and in particular rising pharmaceutical costs, as well the growing societal discussion on mitigating healthcare expenditure. Retail drug prices display a high degree of variance, both within and between countries. This systematic review revealed empirical evidence of determinants retailing to a country's income level, its pharmaceutical market system and its policies and government. Policymakers may be particularly interested in the policy-related determinants of lower retail prices. Drug prices in low-income countries are consistently higher after adjustment for PPP, indicating that lower-income countries still pay the highest relative retail prices. Various easily implementable governmental interventions and regulations, as well as modifiable market



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system-related characteristics, are consistently associated with lower retail prices and thus potential price reduction, such as increasing the generic portions of the drug market, governmental policies such as tendering, central purchasing and regulation schemes targeting annual price inflation, patent regulation and mark-ups, which might aid developing countries in closing the gap of purchasing power-adjusted prices with high-income countries in the future. Most prominently, national and international net price studies are necessary to quantify the effectiveness of these policy measures on net drug prices. We call for openness of net drug price data, as this is our key to change the healthcare cost burden.

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All used data are published and available online.

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## Ethics approval statement

This article is a systematic review of previously published peer-reviewed literature. It does not concern human research and does not use or generate data regarding characteristics or state of health of individuals.

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## Contributor statement

All authors contributed to the conception and design of the research. JMJD and AdA analyzed the data and drafted the first manuscript version. BvdB and MVH provided in multiple sessions conceptual and detailed feedback for every version of the manuscript.

## Competing interests

The authors have no conflicts of interest to declare.

## Data sharing statement

All data relevant to this study are included in the article or available as supplementary material.

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## Table 1: Study overview

| | Bulfone [53] | Cameron [32] | Clarke [54] | Cuomo [31] | Goldstein [55] | Van Harten [5] | Hill [56] | Iyengar [28] | Leopold [26] | Mahlich [57] | Manova [58] | Muzumdar [25] | Roughead [59] | Roughead [37] | Salmasi [29] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | 2009 | 2009 | 2010 | 2017 | 2017 | 2016 | 2015 | 2016 | 2013 | 2019 | 2018 | 2020 | 2007 | 2018 | 2017 |
| **Design** | Cross-sectional | Cross-sectional | **Longitudinal** | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | **Cross-sectional** | **Longitudinal** | Cross-sectional |
| **Setting** | International | International | International | International | International | International | International | International | International | International | International | International | International | International | International |
| **Countries** | High-income (Australia and US) | Low and middle-income countries | High-income (Australia and UK) | Low and middle-income countries | Middle and high-income | High-income | High and middle-income | High, middle and low-income countries | High-income | High-income | High-income (Europe) | High-income (Australia and US) | High-income (Australia and US) | High-income | High, middle and low-income |
| **Prices** | Retail prices | Government prices | Annual wholesale prices | Buyer prices | Retail and discounted prices | Net drug prices | Lowest available price *and* target price | Ex-factory prices | Ex-factory price | Retail prices | Manufacturer and retail prices | Manufacturer prices | Government price and ex-factory price | Retail prices | Retail prices |
| **Drugs** | Generics | Essential generics and originator | Statins | Essential cancer | Patented cancer drugs | Cancer drugs (monocloncal antibodies) | TKIs with patent expiry < 5 yrs | Hepatitis C | Miscellaneous on-patent drugs | Abirateron | Rheumatoid arthritis biologicals and biosimilars | Etanercept, adalimumab, certolizumab, ustekinumab | On-patent 'significant advancement' drugs | Atorvastatin (generic) | Originator anti-cancer drugs |
| **Study quality** | Fair | Good | Fair | Good | Good | Poor | Fair | Good | Good | Good | Good | Fair | Good | Good | Fair |
| **Association determinant and outcome** | **Transparency**: - (qualitative) **Discount**: - (qualitative) | **Mark-up:** + (high in generics) **GDP:** NS (low vs. middle), **Geographic** Africa + **Government** tax on generics: + **Originator:** + (although smaller mark-ups) | **Life cycle:** fraction of generics -, patent expiry - | **GDP:** NS (low vs. middle) **Geographic:** Africa + **Life cycle**: NS for time passing, introduction of generics - | **GDP:** NS (high vs. middle, PPP-adjusted -) | **Governmental**: central purchasing: NS, France - **Geographic:** GDP NS (high-income) **Discounts:** - (Spain) | **Geographic**: US + (vs. Europe), India - (vs middle-income), high vs. middle-income - **Availability of generics**: - | **Government:** tiered pricing agreements - **GDP**: + (low vs. middle), NS (high vs. middle) **Geographic:** US + (unadjusted), Eastern Europe + (adjusted) | **Geographic:** West-/Northern Europe + (vs. Southern) Recession - **Life cycle:** NS over time | **Geographic:** NS | **Life cycle**: Patent expiry - | **Geographic**: US+ (vs. Australia), **Life cycle**: longitudinal: US+, Australia - | **Government**: Government buying scheme - **Geographic**: US and Australia NS | **Policy:** Tendering -, regulation scheme -, patent expiry -, | **GDP:** + (high-income vs. middle and low), NS (high vs. high-middle income), - (low-income vs. high and middle income) |

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| | Savage [24] | Schlenker [23] | Simoens [39] | Srivastava [60] | Suh [30] | Vögler [36] | Vögler [11] | Vögler [10] | Vögler [50] | Wouters [27] | Gong [61] | Kolasani [62] | Ongarora [33] | Scott [34] | Theisen [35] | Vögler [63] |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **Year** | 2017 | 2015 | 2007 | 2014 | 2009 | 2012 | 2016 | 2015 | 2017 | 2017 | 2017 | 2016 | 2019 | 2019 | 2019 | 2013 |
| **Design** | **Longitudinal** | **Longitudinal** | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional | Cross-sectional |
| **Setting** | International | International | International | International | International | International | International | International | International | International | National (US) and International | National | National | National | National | National |
| **Countries** | High-income (US and UK) | High-income | High-income and India (middle) | Low-income | High, middle and low-income | High-income | High-income | High-income | High-income | High-income | High-income | Low-middle income (India) | Kenya | High-income (US) | United States | Austria (high-income) |
| **Prices** | Retail prices | Retail prices | Ex-factory price | Government price | Retail prices | Ex-factory prices | Ex-factory price | Ex-factory price | Ex-factory price | Ex-factory and retail price | Wholesale, commercial, governmental health plan prices | Retail prices | Retail price | Out-of-pocket price | Out-of-pocket price | Retail prices |
| **Drugs** | High-volume anti-cancer drugs | Glaucoma medication | Generic medications in ambulatory care | Miscellaneous generic and originator drugs | Alzheimer's medications | Miscellaneous generic and originator drugs | Originator cancer drugs | Miscellaneous generic and originator drugs | High-cost originator drugs | Miscellaneous generic drugs | Ophthalmology | Cancer drugs | Miscellaneous generic and originator drugs | Ophthalmic | Generic BPH medications | Miscellaneous generic and originator drugs |
| **Study quality** | Poor | Poor | Poor | Good | Fair | Fair | Fair | Fair | Fair | Fair | Fair | Poor | Fair | Fair | Fair | Fair |
| **Association determinant and outcome** | **Government**: price regulation scheme - **Geographic**: US + (vs. UK) **Life cycle**: US+ | **Government**: EPR-, price coupling to consumer price index **Geographic:** US + (vs. Canada), **Longitudinal price development**: US+, Canada -. | **Geographic**: Mature generics market: -, India - **Free market**: + | **Mark-ups +, GDP** NS, **Life cycle**: originator variance ++ (vs. generic) | **GDP**: + (only unadjusted) | **Government:** No generic pricing policy, variance + Frequent tendering - Reference pricing - Free market, variance + **Geographic:** Scandinavian, variance + Size of generics market NS | **GDP**: + (high-income) **Life cycle**: *generics availability* - | **GDP**: + (high-income) **Life cycle**: *generics availability* - | **GDP:** + (high-income) | **Government**: *generics tendering* - **Geographic:** Northern Europe - (vs. Southern) | **Governmental health plan: - Geographic**: US + **Originator: +** | **Life cycle:** number of manufacturers NS, patent + (variance & price) | High variance in and between originator and generics | **Geography**: NS **Type of pharmacy** NS | **Geographic:** high national variance, type of pharmacy: independency - (chain>wholesale>independent) | **Discounts:** - (in-hospital, less expensive medicines) **Life cycle:** On-patent often no discount **Mark-ups**: + |

*Price category listed as mentioned or as described in the article.*
BPH = benign prostate hypertrophy
GDP = gross domestic product
NS = non-significant outcome
PPP = purchasing power parity (*adjusted or unadjusted for PPP*)

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US = United States

**Table 2: Best-evidence synthesis**

| Category | Determinants | Studies | Effect | References | Level of evidence |
|---|---|---|---|---|---|
| Non-modifiable (Geographic) | **Regional/location** | | | | |
| | *GDP (high vs middle-income)* | 2 HQ, 3 FQ | NS, NS, +, NS, + | [28-30, 55, 56] | **Moderate for no association** |
| | *GDP (middle vs. low-income)* | 3 HQ, 2 FQ | NS, NS, **+, +,** + | [28-32] | **Moderate for positive association** |
| | *Europe (west/north vs. south/east)* | 2 HQ, 1 FQ | +, **-, -** | [26-28] | Conflicting evidence |
| | *United States (vs. other high-income)* | 2 HQ, 3 FQ, 1 LQ | NS, **+, +, +, +,+,+** | [23-25, 28, 56, 59, 61] | **Strong for positive association** |
| Modifiable | **Market characteristics** | | | | |
| | *High originator proportion* | 1 HQ, 1 FQ, 1 LQ | **+, +,** + | [31, 32, 61] | **Strong for positive association** |
| | *High generics proportion* | 4 FQ, 1 HQ, 1 LQ | **-, -, -,** NS**, -, -** | [10, 11, 36, 39, 54, 56] | **Strong for negative association** |
| | **Drug life cycle** | | | | |
| | *Time-longitudinal price development* | 2 HQ, 2 LQ | NS, NS, +, + | [23, 24, 26, 31] | Conflicting evidence |
| | *Patent expiry* | 2 HQ, 1 FQ | **-, -, -** | [37, 54, 58] | **Strong for negative association** |
| | **Discounts** | 2 FQ, 1 LQ | **-, -,** NS | [5, 53, 63] | **Moderate for negative association** |
| | **Mark-up** | 2 HQ, 1 FQ | **+, +,** + | [32, 60, 63] | **Strong for positive association** |
| Modifiable | **Policy** | | | | |
| | *Tendering* | 1 HQ, 2 FQ | **-, -, -** | [27, 36, 37] | **Strong for negative association** |
| | *Central purchaser* | 1 HQ, 1 FQ, 1 LQ | **-, -,** NS | [5, 36, 59] | **Moderate for negative association** |
| | *Pricing regulation (indexation) scheme* | 1 HQ, 2 LQ | **-, -, -** | [23, 24, 37] | **Moderate for negative association** |

**Abbreviations**

GDP: gross domestic product, HQ: high quality, FQ: fair quality, LQ: low quality, NS: not significant, + : positive effect, - : negative effect

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# References


1. Keehan, S.P., et al., *National health expenditure projections, 2014-24: spending growth faster than recent trends.* Health Aff (Millwood), 2015. **34**(8): p. 1407-17.
2. Murthy, V.N.R. and A.A. Okunade, *Determinants of U.S. health expenditure: Evidence from autoregressive distributed lag (ARDL) approach to cointegration.* Economic Modelling, 2016. **59**: p. 67-73.
3. *The Global Use of Medicine in 2019 and Outlook to 2013: Forecasts and Areas to Watch.* IQVIA Institute for Human Data Science, 2019.
4. Hernandez, I., et al., *Changes in List Prices, Net Prices, and Discounts for Branded Drugs in the US, 2007-2018.* JAMA, 2020. **323**(9): p. 854-862.
5. van Harten, W.H., et al., *Actual costs of cancer drugs in 15 European countries.* Lancet Oncol, 2016. **17**(1): p. 18-20.
6. De Block, M., *The difficulty of comparing drug prices between countries.* Lancet Oncol, 2016. **17**(4): p. e125.
7. Dana, K.N., J.B. Hertig, and R.J. Weber, *Drug Pricing Transparency: The New Retail Revolution.* Hosp Pharm, 2017. **52**(2): p. 155-159.
8. Rintoul, A., et al., *Medicines with one seller and many buyers: strategies to increase the power of the payer.* BMJ, 2020. **369**: p. m1705.
9. Shrader, K.R., R.J. DeSalvo, and R.V. Evanson, *A comparison study of drug prices in segmented markets.* J Am Pharm Assoc, 1971. **11**(11): p. 598-602.
10. Vogler, S., K. Kilpatrick, and Z.U. Babar, *Analysis of Medicine Prices in New Zealand and 16 European Countries.* Value Health, 2015. **18**(4): p. 484-92.
11. Vogler, S., A. Vitry, and Z.U. Babar, *Cancer drugs in 16 European countries, Australia, and New Zealand: a cross-country price comparison study.* Lancet Oncol, 2016. **17**(1): p. 39-47.
12. Vogler, S., et al., *How Can Pricing and Reimbursement Policies Improve Affordable Access to Medicines? Lessons Learned from European Countries.* Appl Health Econ Health Policy, 2017. **15**(3): p. 307-321.
13. Ferrario, A., et al., *Strategies to achieve fairer prices for generic and biosimilar medicines.* BMJ, 2020. **368**: p. l5444.
14. Kanavos, P.G. and S. Vandoros, *Determinants of branded prescription medicine prices in OECD countries.* Health Econ Policy Law, 2011. **6**(3): p. 337-67.
15. Hanlon, M. and R. Zhang, *Price discrimination in essential medicines: evidence from International Drug Price Indicator Guide data.* Int Health, 2013. **5**(1): p. 58-63.
16. Russo, G. and B. McPake, *Medicine prices in urban Mozambique: a public health and economic study of pharmaceutical markets and price determinants in low-income settings.* Health Policy Plan, 2010. **25**(1): p. 70-84.
17. Yao, Y. and M. Tanaka, *Price offers of pharmaceutical procurement in China: evidence from Guangdong province.* Eur J Health Econ, 2016. **17**(5): p. 563-75.
18. Dylst, P. and S. Simoens, *Does the market share of generic medicines influence the price level?: a European analysis.* Pharmacoeconomics, 2011. **29**(10): p. 875-82.
19. Puig-Junoy, J., *Impact of European pharmaceutical price regulation on generic price competition: a review.* Pharmacoeconomics, 2010. **28**(8): p. 649-63.
20. Moher, D., et al., *Preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement.* BMJ, 2009. **339**: p. b2535.
21. *WHO Guideline on Country Pharmaceutical Pricing Policies* World Health Organization, 2015.
22. Health, N.I.o., *Quality Assessment Tool for Observational Cohort and Cross-Sectional Studies.* Study Quality Assessment Tools, 2020.
23. Schlenker, M.B., G.E. Trope, and Y.M. Buys, *Comparison of United States and canadian glaucoma medication costs and price change from 2006 to 2013.* J Ophthalmol, 2015. **2015**: p. 547960.
24. Savage, P., et al., *Cancer Drugs: An International Comparison of Postlicensing Price Inflation.* J Oncol Pract, 2017. **13**(6): p. e538-e542.
25. Muzumdar, S., R. Wu, and H. Feng, *Temporal comparison of biologic medication prices between the United States and Australia.* J Dermatolog Treat, 2020: p. 1-2.
26. Leopold, C., et al., *Is Europe still heading to a common price level for on-patent medicines? An exploratory study among 15 Western European countries.* Health Policy, 2013. **112**(3): p. 209-16.


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27. Wouters, O.J. and P.G. Kanavos, *A comparison of generic drug prices in seven European countries: a methodological analysis.* BMC Health Serv Res, 2017. **17**(1): p. 242.
28. Iyengar, S., et al., *Prices, Costs, and Affordability of New Medicines for Hepatitis C in 30 Countries: An Economic Analysis.* PLoS Med, 2016. **13**(5): p. e1002032.
29. Salmasi, S., et al., *Pricing appraisal of anti-cancer drugs in the South East Asian, Western Pacific and East Mediterranean Region.* BMC Cancer, 2017. **17**(1): p. 903.
30. Suh, G.H., et al., *International price comparisons of Alzheimer's drugs: a way to close the affordability gap.* Int Psychogeriatr, 2009. **21**(6): p. 1116-26.
31. Cuomo, R.E., R.L. Seidman, and T.K. Mackey, *Country and regional variations in purchase prices for essential cancer medications.* BMC Cancer, 2017. **17**(1): p. 566.
32. Cameron, A., et al., *Medicine prices, availability, and affordability in 36 developing and middle-income countries: a secondary analysis.* Lancet, 2009. **373**(9659): p. 240-9.
33. Ongarora, D., et al., *Medicine Prices, Availability, and Affordability in Private Health Facilities in Low-Income Settlements in Nairobi County, Kenya.* Pharmacy (Basel), 2019. **7**(2).
34. Scott, A.T., P.E. Pecen, and A.G. Palestine, *Ophthalmic medication price variation across the United States: Anti-inflammatory medications.* Ther Adv Ophthalmol, 2019. **11**: p. 2515841419863638.
35. Theisen, K.M., et al., *Extreme Price Variation for Generic Benign Prostatic Hyperplasia Medications.* Urology, 2019. **124**: p. 223-228.
36. S., V., *How large are the differences between originator and generic prices? Analysis of five molecules in 16 European countries.* Economics, 2012. **DOI:10.7175/fe.v13i3S.311**.
37. Roughead, E.E., et al., *Pricing policies for generic medicines in Australia, New Zealand, the Republic of Korea and Singapore: patent expiry and influence on atorvastatin price.* WHO South East Asia J Public Health, 2018. **7**(2): p. 99-106.
38. Howard, D.H., et al., *Pricing in the Market for Anticancer Drugs.* J Econ Perspect, 2015. **29**(1): p. 139-62.
39. Simoens, S., *International comparison of generic medicine prices.* Curr Med Res Opin, 2007. **23**(11): p. 2647-54.
40. Sarpatwari, A., et al., *Competition and price among brand-name drugs in the same class: A systematic review of the evidence.* PLoS Med, 2019. **16**(7): p. e1002872.
41. Aalto-Setala, V., *The impact of generic substitution on price competition in Finland.* Eur J Health Econ, 2008. **9**(2): p. 185-91.
42. Maniadakis, N., et al., *Shaping Pharmaceutical Tenders for Effectiveness and Sustainability in Countries with Expanding Healthcare Coverage.* Appl Health Econ Health Policy, 2018. **16**(5): p. 591-607.
43. Sood N, d.V.H., Gutierrez Italo, Lakdawalla D, Goldman DP, *The Effect Of Regulation On Pharmaceutical Revenues: Experience In Nineteen Countries.* Health Affairs, 2008. **27**(Supplement 1): p. w125-w137.
44. Lakdawalla, D., *Economics of the Pharmaceutical Industry.* Journal of Economic Literature 2018, 56(2), 397–449, 2018.
45. *World Medicines Situation - Medicines Expenditures.* World Health Organization, 2011.
46. Maniadakis, N., et al., *Comprehensive taxonomy and worldwide trends in pharmaceutical policies in relation to country income status.* BMC Health Serv Res, 2017. **17**(1): p. 371.
47. Zaidi, S., et al., *Access to essential medicines in Pakistan: policy and health systems research concerns.* PLoS One, 2013. **8**(5): p. e63515.
48. European Parliament, D.-g.f.i.p., *Differences in Costs of and Access to Pharmaceutical Products in the EU.* 2011.
49. Borges Dos Santos, M.A., et al., *Factors influencing pharmaceutical pricing - a scoping review of academic literature in health science.* J Pharm Policy Pract, 2019. **12**: p. 24.
50. Vogler, S., N. Zimmermann, and Z.U. Babar, *Price comparison of high-cost originator medicines in European countries.* Expert Rev Pharmacoecon Outcomes Res, 2017. **17**(2): p. 221-230.
51. Vogler, S., P. Schneider, and N. Zimmermann, *Evolution of Average European Medicine Prices: Implications for the Methodology of External Price Referencing.* Pharmacoecon Open, 2019. **3**(3): p. 303-309.
52. WHO, *World Health Organization: Medicine price information sources.* https://www.who.int/medicines/areas/access/med_price_info_sources/en/, last accessed February 3rd, 2021.
53. Bulfone, L., *High prices for generics in Australia - more competition might help.* Aust Health Rev, 2009. **33**(2): p. 200-14.


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54. Clarke, P.M. and E.M. Fitzgerald, *Expiry of patent protection on statins: effects on pharmaceutical expenditure in Australia.* Med J Aust, 2010. **192**(11): p. 633-6.
55. Goldstein, D.A., et al., *A global comparison of the cost of patented cancer drugs in relation to global differences in wealth.* Oncotarget, 2017. **8**(42): p. 71548-71555.
56. Hill, A., et al., *Target prices for mass production of tyrosine kinase inhibitors for global cancer treatment.* BMJ Open, 2016. **6**(1): p. e009586.
57. Mahlich, J., J. Sindern, and M. Suppliet, *Cross-national drug price comparisons with economic weights in external reference pricing in Germany.* Expert Rev Pharmacoecon Outcomes Res, 2019. **19**(1): p. 37-43.
58. Manova, M., et al., *Comparative Price Analysis of Biological Products for Treatment of Rheumatoid Arthritis.* Front Pharmacol, 2018. **9**: p. 1070.
59. Roughead, E.E., R. Lopert, and L.N. Sansom, *Prices for innovative pharmaceutical products that provide health gain: a comparison between Australia and the United States.* Value Health, 2007. **10**(6): p. 514-20.
60. Srivastava, D. and A. McGuire, *Analysis of prices paid by low-income countries - how price sensitive is government demand for medicines?* BMC Public Health, 2014. **14**: p. 767.
61. Gong, D., et al., *Comparison of United States and International Ophthalmic Drug Pricing.* Ophthalmology, 2019. **126**(10): p. 1358-1365.
62. Kolasani, B.P., D.C. Malathi, and R.R. Ponnaluri, *Variation of Cost among Anti-cancer Drugs Available in Indian Market.* J Clin Diagn Res, 2016. **10**(11): p. FC17-FC20.
63. Vogler, S., et al., *The role of discounts and loss leaders in medicine procurement in Austrian hospitals - a primary survey of official and actual medicine prices.* Cost Eff Resour Alloc, 2013. **11**(1): p. 15.

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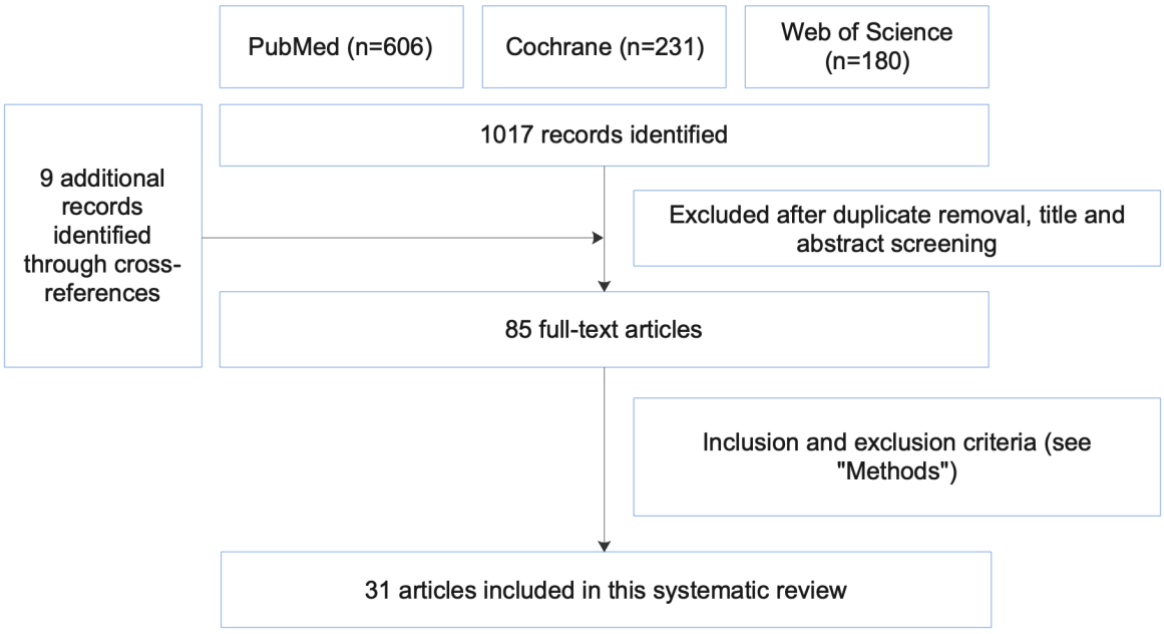
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# Supplementary materials

**eTable 1: PubMed Search Strategy**

(((((((("Pharmaceutical Preparations"[Mesh]) OR ((medicine[Title/Abstract] OR medicines[Title/Abstract] OR drug[Title/Abstract] OR drugs[Title/Abstract] OR pharmaceutical[Title/Abstract] OR pharmaceuticals[Title/Abstract] OR medication[Title/Abstract] OR medications[Title/Abstract])))) AND ((price[Title/Abstract] OR prices[Title/Abstract] OR pricing[Title/Abstract])))) AND (("Benchmarking"[Mesh]) OR ((benchmark[Title/Abstract] OR compare[Title/Abstract] OR comparison[Title/Abstract] OR comparisons[Title/Abstract])))) NOT cost-effectiveness AND (full text[sb])

**eFigure 1: Study selection process**




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**eTable 2: Other determinants not in best-evidence synthesis (< 3 records)**

Supplementary

| Category | Determinant | Studies | Effect | References |
|---|---|---|---|---|
| **Geography** | *Africa (vs. low-income)* | 2 HQ | +, + | [31, 32] |
| | *France* | 1 LQ | + | [5] |
| | *India* | 2 FQ | -,- | [39, 56] |
| **(Pharmaceutical) market** | *Number of manufacturers* | 1 LQ | NS | [62] |
| | *Tiered pricing agreement* | 1 HQ | - | [28] |
| | *Independency of pharmacy* | 2 FQ | -, NS | [34, 35] |
| | *Recession* | 1 HQ | - | [26] |
| **Policy** | *Governmental health plan* | 1 FQ | - | [61] |
| | *EPR* | 1 LQ, 1 FQ | -, - | [23, 36] |
| | *Transparency* | 1 FQ | - | [53] |
| | *Free market* | 1 LQ | + | [39] |
| | *Generics tax* | 1 HQ | + | [32] |

HQ: high quality, FQ: fair quality, LQ: low quality, NS: not significant, + : positive effect, - : negative effect, EPR: external price referencing

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# PRISMA 2009 Checklist

| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| **TITLE** | | | |
| Title | 1 | Identify the report as a systematic review, meta-analysis, or both. | **1** |
| **ABSTRACT** | | | |
| Structured summary | 2 | Provide a structured summary including, as applicable: background; objectives; data sources; study eligibility criteria, participants, and interventions; study appraisal and synthesis methods; results; limitations; conclusions and implications of key findings; systematic review registration number. | **2** |
| **INTRODUCTION** | | | |
| Rationale | 3 | Describe the rationale for the review in the context of what is already known. | **3** |
| Objectives | 4 | Provide an explicit statement of questions being addressed with reference to participants, interventions, comparisons, outcomes, and study design (PICOS). | **3** |
| **METHODS** | | | |
| Protocol and registration | 5 | Indicate if a review protocol exists, if and where it can be accessed (e.g., Web address), and, if available, provide registration information including registration number. | **-** |
| Eligibility criteria | 6 | Specify study characteristics (e.g., PICOS, length of follow-up) and report characteristics (e.g., years considered, language, publication status) used as criteria for eligibility, giving rationale. | **4** |
| Information sources | 7 | Describe all information sources (e.g., databases with dates of coverage, contact with study authors to identify additional studies) in the search and date last searched. | **4** |
| Search | 8 | Present full electronic search strategy for at least one database, including any limits used, such that it could be repeated. | **4-+17** |
| Study selection | 9 | State the process for selecting studies (i.e., screening, eligibility, included in systematic review, and, if applicable, included in the meta-analysis). | **4** |
| Data collection process | 10 | Describe method of data extraction from reports (e.g., piloted forms, independently, in duplicate) and any processes for obtaining and confirming data from investigators. | **4-5** |
| Data items | 11 | List and define all variables for which data were sought (e.g., PICOS, funding sources) and any assumptions and simplifications made. | **4-5** |
| Risk of bias in individual studies | 12 | Describe methods used for assessing risk of bias of individual studies (including specification of whether this was done at the study or outcome level), and how this information is to be used in any data synthesis. | **5** |
| Summary measures | 13 | State the principal summary measures (e.g., risk ratio, difference in means). | **5** |
| Synthesis of results | 14 | Describe the methods of handling data and combining results of studies, if done, including measures of consistency (e.g., $I^2$) for each meta-analysis. | **5** |

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# PRISMA 2009 Checklist



| Section/topic | # | Checklist item | Reported on page # |
|---|---|---|---|
| Risk of bias across studies | 15 | Specify any assessment of risk of bias that may affect the cumulative evidence (e.g., publication bias, selective reporting within studies). | **5** |
| Additional analyses | 16 | Describe methods of additional analyses (e.g., sensitivity or subgroup analyses, meta-regression), if done, indicating which were pre-specified. | **5** |
| **RESULTS** | | | |
| Study selection | 17 | Give numbers of studies screened, assessed for eligibility, and included in the review, with reasons for exclusions at each stage, ideally with a flow diagram. | **5-6** |
| Study characteristics | 18 | For each study, present characteristics for which data were extracted (e.g., study size, PICOS, follow-up period) and provide the citations. | **6** |
| Risk of bias within studies | 19 | Present data on risk of bias of each study and, if available, any outcome level assessment (see item 12). | **9-10** |
| Results of individual studies | 20 | For all outcomes considered (benefits or harms), present, for each study: (a) simple summary data for each intervention group (b) effect estimates and confidence intervals, ideally with a forest plot. | **6-7-15-16** |
| Synthesis of results | 21 | Present results of each meta-analysis done, including confidence intervals and measures of consistency. | **6-7** |
| Risk of bias across studies | 22 | Present results of any assessment of risk of bias across studies (see Item 15). | **9-10** |
| Additional analysis | 23 | Give results of additional analyses, if done (e.g., sensitivity or subgroup analyses, meta-regression [see Item 16]). | **17** |
| **DISCUSSION** | | | |
| Summary of evidence | 24 | Summarize the main findings including the strength of evidence for each main outcome; consider their relevance to key groups (e.g., healthcare providers, users, and policy makers). | **7-8-9** |
| Limitations | 25 | Discuss limitations at study and outcome level (e.g., risk of bias), and at review-level (e.g., incomplete retrieval of identified research, reporting bias). | **9-10** |
| Conclusions | 26 | Provide a general interpretation of the results in the context of other evidence, and implications for future research. | **10** |
| **FUNDING** | | | |
| Funding | 27 | Describe sources of funding for the systematic review and other support (e.g., supply of data); role of funders for the systematic review. | **11** |

*From:* Moher D, Liberati A, Tetzlaff J, Altman DG, The PRISMA Group (2009). Preferred Reporting Items for Systematic Reviews and Meta-Analyses: The PRISMA Statement. PLoS Med 6(7): e1000097. doi:10.1371/journal.pmed1000097

For more information, visit: **www.prisma-statement.org**.



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